Rel-Sortability of Time-Unrolled Causal Graphs: Exact Values, Finite-Sample Saturation, and Real-Data Tests

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Abstract

Causal-discovery benchmarks can often be solved by sorting variables by a marginal statistic. Rel-sortability orders nodes by their number of relatives and was shown to tend to the uninformative value $1/2$ on time-unrolled DAGs whose summary graph is strongly connected in every weakly connected component. We give the exact rel-sortability of every edge of the time-unrolled DAG, for every summary graph with self-loops and every horizon, with an $O(1/T)$ rate, and show that the condition is also necessary: every other summary graph, for instance any river network, stays strictly above $1/2$. Much of this excess is an artefact of the fixed start of the simulation. With a stationary start the value is exact for every horizon and depends only on the source strongly connected components. For Erdős–Rényi summary graphs we prove the limiting values on paper. The graph results are machine-checked in Lean 4. A finite-sample analysis says when a relatives count can be estimated from one autocorrelated record and when a latent common driver saturates it. In real-data studies with internally time-stamped analysis plans, a river benchmark failed (a post hoc analysis is consistent with saturation) and a physical testbed gave partial support. In a benchmark experiment with time labels hidden, the fixed start raised the accuracy of a relatives-sorting baseline by amounts that track the exact values and reversed several method rankings.

1. Introduction

Simulated benchmarks drive much of causal discovery, and several of their regularities can be exploited without any causal reasoning. Marginal variances tend to grow along the causal order (*var-sortability*; Reisach et al., 2021), so a method that sorts by variance does well on data from common generators, and it explains the strong benchmark performance of some continuous-optimisation methods, a family that includes NOTEARS (Zheng et al., 2018). The coefficient of determination R^2 has the same property and is scale invariant (Reisach et al., 2023); internally standardised structural models are not var-sortable by construction and, empirically, mostly not R^2-sortable (Ormaniec et al., 2025); and Herman et al. (2025) relate the R^2 pattern to colliders and confounders and propose a generator that avoids it. Most recently, Reisach et al. (2026) defined *rel-sortability*, which orders a node by the number of its *relatives* (nodes joined to it by an open path, marginally dependent on it under faithfulness). Relatives are a property of the graph alone, so the criterion cannot be removed by reweighting. For random DAGs it is high; for *time-unrolled* DAGs built from a summary graph with self-loops, Reisach et al. (2026, Thm. 4) show that it tends to $1/2$ if every weakly connected component of the summary graph is strongly connected, and they propose such graphs as benchmarks. Only sufficiency is shown in the arXiv version; they do not analyse the value beyond this case.

Three questions remain open, and they matter for real time series, whose summary graphs are often feed-forward (rivers, supply chains, signalling cascades): (Q1) what is the rel-sortability of a time-unrolled DAG when the summary graph is not strongly connected; (Q2) does it depend on how the series is started and observed; and (Q3) can a relatives count be estimated from one finite, autocorrelated record, so that a method could actually exploit it?

Contributions. **(1)** The exact rel-sortability of every edge of the time-unrolled DAG G_T, for every summary graph with self-loops and every T (Theorem 1), its limit $\frac{1}{2} + |E_\times|/(2|E_S|)$ with an explicit $O(1/T)$ rate (Theorem 2), and necessity of the condition of Reisach et al. (2026, Thm. 4) (Corollary 3). The excess above $1/2$ for chains and out-trees is a boundary effect of the deterministic start (Proposition 4); with a stationary start the value is exact for every T and fixed by the *source* strongly connected components (Theorem 5 and Corollary 6). **(2)** Limits for

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directed Erdős–Rényi summary graphs, proved on paper (Proposition 7, Appendix N.1). **(3)** A finite-sample layer (Section 5): an effective sample size and recovery bound for the relatives count, saturation by a latent driver at $n \approx z_c^2/\ell^2$, and conditions for a proxy to repair it. **(4)** A Lean 4 formalisation of the exact and stationary results (Section 6), and real-data studies with time-stamped analysis plans (Section 7), reported with every failed test. **(5)** A benchmark experiment (Section 3.2): with time labels hidden, the fixed start shifts sort-based baselines by amounts the exact values predict and reverses significant pairwise rankings.

The graph results of (1) are short consequences of a walk-length lemma and of Reisach et al. (2026, Lemma 1). Their value is that they determine the value for every summary graph, horizon and start, and change what a benchmark designer should conclude: much of the rel-sortability of time-unrolled benchmarks is an artefact of how the simulation is started. The finite-sample results use textbook tools. We see the paper as a completion of the unrolled-graph analysis of Reisach et al. (2026, Section 5) plus a finite-sample and real-data layer, not as a new theory of sortability.

Related work. Varsortability (Reisach et al., 2021) and R^2-sortability (Reisach et al., 2023) have been analysed for standardised SCMs (Ormaniec et al., 2025), for unit-free generators with a time-series extension (Herman et al., 2025), and, for time series, by Lohse & Wahl (2025), who adapt both criteria to summary graphs, note that cyclic pairs push the signal towards $1/2$, and measure them on river-flow and Causal Chambers data; they give no theorem and do not study relatives or observation schemes. Reisach et al. (2026) introduce rel-sortability and the time-unrolled DAG, and prove that strong connectivity of every weak component of the summary graph is sufficient for the limit $1/2$. Block means of a VAR converge to a static linear SEM with coefficients $A_{ji}/(1 - A_{jj})$ (Gong et al., 2017); that work and Gong et al. (2015) study identifiability from aggregated or subsampled data, not sortability, and aggregation is classically known to change time-series models (Weiss, 1984) and to induce contemporaneous correlation among the innovations of the aggregated system (Breitung & Swanson, 2002). The size distortion of a Pearson test with the nominal sample size under autocorrelation is classical (Bartlett, 1935; Clifford et al., 1989; Roy, 1989) and is handled in practice by effective sample sizes (Runge et al., 2019; Afyouni et al., 2019); testing the independence of two series through lagged cross-correlations goes back to Haugh (1976). Residual confounding by unreliable covariates is also known (Westfall & Yarkoni, 2016). Formal libraries for causal inference exist (Tan & Syrgkanis, 2026). Time-series discovery methods such as PCMCI (Runge et al., 2019) test conditional independences, whereas our finite-sample results concern only the marginal tests that define relatives.

2. Setting

For a directed graph, $\mathrm{anc}(y)$ and $\mathrm{desc}(y)$ contain y; the relatives of y are $\mathrm{rel}(y) = \mathrm{desc}(\mathrm{anc}(y)) = \{z : \mathrm{anc}(y) \cap \mathrm{anc}(z) \neq \emptyset\}$ and $\rho(y) = |\mathrm{rel}(y)|$. For an edge (x, y), $s_\rho(x, y) = 1, \frac{1}{2}, 0$ if $\rho(x) <, =, > \rho(y)$, and $s_\rho(G)$ is the mean over *edges*, as in Reisach et al. (2026, Eq. (4)) (quoted verbatim in Appendix A). A value $1/2$ means the criterion carries no orientation information.

A *summary graph* $G_\mathrm{S} = (V, E_\mathrm{S})$ has $p \geq 2$ nodes and a self-loop at every node. The *time-unrolled DAG* G_T has nodes $v(t)$, $t \in \{1, \ldots, T\}$, and an edge $(i(t), j(t{+}1))$ for each $(i, j) \in E_\mathrm{S}$ and $t < T$, so every summary edge has $T - 1$ copies, all at lag one (a VAR(1)-shaped unrolling; other lags are discussed after Theorem 5). $d(u, v)$ is the shortest-path length in G_S (self-loops ignored; $d(v, v) = 0$; ∞ if unreachable), and anc_S, rel_S are taken in G_S. SCC means strongly connected component. $E_\times \subseteq E_\mathrm{S}$ is the set of edges between different SCCs and $E_\circ = E_\mathrm{S} \setminus E_\times$. A *source* SCC has no incoming edge from another SCC, and $\mathrm{Src}(v)$ is the set of source SCCs that reach v.

3. Exact Rel-Sortability of Time-Unrolled DAGs

Theorem 1 (every edge, every T)**.** *For $(i, j) \in E_\mathrm{S}$ and $1 \leq t < T$, the edge $(i(t), j(t{+}1))$ of G_T has $s_\rho = 1$ if some $u \in V$ satisfies $d(u, j) \leq t \leq d(u, i)$, and $s_\rho = 1/2$ otherwise. The value does not depend on T.*

Lean: `edge_score_dist`, `edge_score`, `strictCond_iff_dist`.

Sketch. Every edge of G_T advances time by one, so $u(s)$ is an ancestor of $v(t)$ iff $d(u, v) \leq t - s$ (pad a shortest path with self-loops). Hence the roots of $v(t)$ are $\{u(1) : d(u, v) \leq t - 1\}$. Since $i(t)$ is a parent of $j(t{+}1)$, $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$, and the inclusion is strict iff $j(t{+}1)$ has a root that $i(t)$ lacks (Reisach et al., 2026, Thm. 1, Lemma 1). Full proof in Appendix A. (The Lean proof characterises rel in G_T directly and does not use their Lemma 1.)

Theorem 2 (exact limit and rate)**.** *For every $T \geq 2$, with $L(G_\mathrm{S}) = \frac{1}{2} + \frac{|E_\times|}{2|E_\mathrm{S}|}$ and $\Delta \leq p - 1$ the largest finite shortest-path distance in G_S,*

$$0 \leq s_\rho(G_T) - L(G_\mathrm{S}) \leq \frac{\min(\Delta, T{-}1)\,|E_\circ|}{2(T-1)|E_\mathrm{S}|} \leq \frac{p-1}{2(T-1)},$$

so $\lim_{T\to\infty} s_\rho(G_T) = L(G_\mathrm{S})$.

Lean: `sRho_unrolled_exact`, `sRho_unrolled_bounds`, `tendsto_sRho_GT`.

Sketch. For a crossing edge, j does not reach i; take $u = j$ in Theorem 1: all $T - 1$ copies score 1. For an edge inside an SCC, i and j are reached by the same nodes, which forces $t \le d(u, i) \le \Delta$, so at most Δ copies score 1. (Lean checks the bound with $p - 1$; the Δ form is this one-line paper argument, checked by enumeration on 21,000 graph–horizon cases with no violation, Appendix P.)

Corollary 3 (necessary and sufficient condition)**.** $\lim_T s_\rho(G_T) = 1/2$ *iff every weakly connected component of* G_S *is strongly connected. Otherwise* $s_\rho(G_T) \ge \frac{1}{2} + \frac{1}{2|E_S|}$ *for every* T*. The "if" half is Reisach et al. (2026, Thm. 4). If* G_S *is acyclic apart from self-loops, with* m *other edges,* $L = \frac{1}{2} + \frac{m}{2(m+p)}$ *whatever its shape (chain, in-tree, out-tree, collider).*

Lean: `tendsto_half_iff`, `crossEdges_eq_empty_tfae`, `sRho_unrolled_ge_of_cross`, `reisach_theorem4`, `limitValue_dag`.

Proposition 4 (the excess is a boundary effect)**.** *For* $t \ge p$, $\rho(v(t)) = T\,|\mathrm{rel}_S(v)| - \sum_{w \in \mathrm{rel}_S(v)} d(\mathrm{anc}_S(v), w)$.

Lean: `rho_unrolled_large_t`, `rho_unrolled_eq`.

So for a crossing edge with $\mathrm{rel}_S(i) = \mathrm{rel}_S(j)$ (root to child in an out-tree, a chain), the strict order rests on $O(1)$ extra relatives out of $\Theta(T)$, all located in the first p slices. Figure 1 shows the smallest case. A sample-based relatives count on a long record cannot see such a difference, which motivates the stationary version.

Theorem 5 (stationary start)**.** *Observe the window of the last* $T \ge 2$ *slices of* G_{B+T} *with a latent burn-in* $B \ge \Delta$ *(or of the infinite-past graph on* $V \times \mathbb{Z}$*), with relatives computed in the full graph. Then every window node has* $\rho_W(v(t)) = T|\mathrm{rel}_S(v)|$*, and for every* T

$$s_\rho^W = \tfrac{1}{2} + \frac{|E_\uparrow|}{2|E_S|},\quad E_\uparrow = \{(i,j) \in E_S : \mathrm{Src}(i) \ne \mathrm{Src}(j)\}.$$

Lean: `rhoBurn_eq`, `sBurn_eq`, `rhoZ_eq`, `sZ_eq`, `relSet_eq_iff_src`.

Sketch. After Δ slices every finite distance is absorbed, so $w(s) \in \mathrm{rel}(v(t))$ iff $\mathrm{anc}_S(v) \cap \mathrm{anc}_S(w) \ne \emptyset$. For an edge (i, j), $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$ with equality iff $\mathrm{Src}(i) = \mathrm{Src}(j)$.

Scope: unit lags. Theorems 1, 2 and 5 are stated, and machine-checked, for VAR(1)-shaped unrollings. If each summary edge (i, j) instead acts at a lag $\tau_{ij} \in \{1, \dots, \tau_{\max}\}$ (self-loops at lag one), the same proofs go through with d replaced by the lag-weighted distance d_τ: the copy $(i(t), j(t+\tau_{ij}))$ is strict iff some u has $d_\tau(u, j) - \tau_{ij} + 1 \le t \le d_\tau(u, i)$, the limit is unchanged, and the stationary result needs a burn-in $B \ge \Delta\tau_{\max}$ (paper proof, Appendix M; not machine-checked). Contemporaneous (lag-0) edges are not covered.

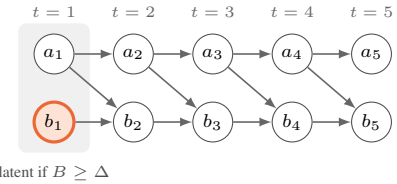


Figure 1. Summary chain $a \to b$ unrolled to $T = 5$. Every a_t has the relatives $\{a_s\} \cup \{b_s : s \ge 2\}$ ($2T - 1$ nodes); every b_t with $t \ge 2$ also has b_1 ($2T$ nodes). So each edge $a_t \to b_{t+1}$ is strictly sorted (finite limit $L = 2/3$), but only because of the root b_1 at the fixed start. With a latent burn-in slice, both counts are $2T$ and $s_\rho^W = 1/2$.

Table 1. Rel-sortability of 3-node summary graphs (with self-loops, $|E_S| = 5$). Finite: $\lim_T s_\rho(G_T)$ (Theorem 2); stationary: s_ρ^W (Theorem 5). Lean: `chain3_limit`, `chain3_stat`, `fork_limit`, `fork_stat`, `collider_limit`, `collider_stat`; 3-cycle by `reisach_theorem4`.

summary graph	finite G_T (limit)	stationary
chain $a \to b \to c$	0.7	0.5
fork $a \leftarrow b \to c$	0.7	0.5
collider $a \to c \leftarrow b$	0.7	0.7
3-cycle (6 edges)	0.5	0.5

Corollary 6 (stationary condition; graph classes)**.** $s_\rho^W = 1/2$ *iff every weakly connected component of* G_S *has exactly one source SCC. Strongly connected graphs, out-forests, chains and forks give* $1/2$*. In an in-forest (a river network),* $s_\rho^W = \frac{1}{2} + \#\{\text{edges into confluences}\}/(2|E_S|)$*, which exceeds* $1/2$ *iff there is a confluence.*

Lean: `sBurn_eq_half_iff`, `sZ_eq_half_iff`, `stat_half_of_outForest`, `statValue_inForest`, `statValue_inForest_gt_iff`, `collider_stat`, `fork_stat`.

Table 1 shows the two versions. In the finite generator every summary DAG with the same m/p has the same value, so a benchmark built this way rewards relatives-sorting on chains and forks, where the stationary process gives no signal; for long stationary records the stationary value is the relevant one. All statements of this section were checked by exhaustive enumeration ($p \le 4$ and 10^4 random graphs with $p \le 7$; 46,724 cases, 0 failures; Appendix B).

Without self-loops (Appendix L). Every edge of G_T still scores $1/2$ or 1, independently of T, by an explicit walk-length criterion (machine-checked), and Reisach et al. (2026, Thm. 4) still holds, but the closed form and the converse of Corollary 3 fail: for the single edge $0 \to 1$ without loops $s_\rho(G_T) = 1/2$ for every T (with loops the limit is $2/3$). The loop assumption is therefore necessary for the closed forms.

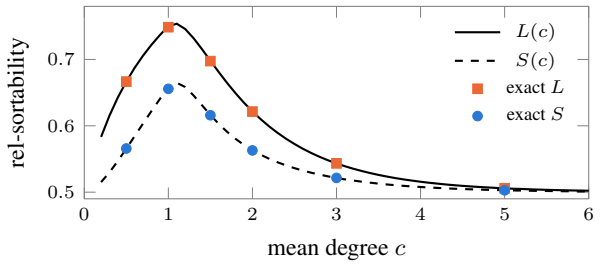


Figure 2. Proposition 7 (lines) against exact computation on seeded graphs ($p = 2{\cdot}10^4$ to 10^5; points, means over seeds). The standard deviations over seeds are at most 0.0022 (Appendix N).

3.1. Random summary graphs

Let G_S be the directed Erdős–Rényi graph $G(p, c/p)$ plus a self-loop at every node, and x the largest root of $x = 1 - e^{-cx}$ ($x = 0$ for $c \le 1$). By Theorems 2 and 5 only $|E_\times|$ and $|E_\uparrow|$ matter.

Proposition 7 (ER limits; proved on paper)**.** *Let $L(G_S) = \lim_{T\to\infty} s_\rho(G_T)$ at fixed p and $S(G_S) = s_\rho^W$. In probability and in L^1 as $p \to \infty$, $L(G_S) \to \frac{1}{2} + \frac{c(1-x^2)}{2(1+c)}$ and $S(G_S) \to \frac{1}{2} + \frac{c\,F_\uparrow}{2(1+c)}$ with $F_\uparrow = (1-x)(1-e^{-c}) + x(1-x)\big(1 - e^{-c(1-x)d}\big)$, where d is the smallest fixed point in $[0,1]$ of $d = 1 + e^{-c} - e^{-c(1-x)d}$ ($d = 1$ for $c \le 1$).*

Lean: only the deterministic witness lemmas for $E_\uparrow$ (`upEdge_witness`, `upEdges_subset_confluenceNL`); the probabilistic proof (Appendix N.1) uses textbook Poisson-coupling, Chernoff and Galton–Watson facts (van der Hofstad, 2016). Not proved: rates, the window $c = 1 + \lambda p^{-1/3}$, a.s. convergence, fixed T as p grows.

The first formula says that the fraction of non-loop edges inside an SCC tends to x^2 (giant strongly connected component, Dorogovtsev et al., 2001; Dovgal et al., 2020). $L(1) = 3/4$, L peaks at 0.754 near $c = 1.07$ and then decays like e^{-c}; S peaks at 0.665 near $c = 1.09$ (Figure 2). Random summary graphs are thus rel-sortable only near or below criticality, as the giant SCC removes the signal. For in-forests (rivers) the identities are exact (Appendix N).

What the value means. Show an edge in random order with the two relatives counts; the rule "orient towards the larger count, flip a coin on ties" has accuracy exactly s_ρ, never errs on an edge it decides, and no pairwise rule on the two counts does better (a few lines from Reisach et al.'s Theorem 1, Appendix O). So $s_\rho = 0.7$ caps such a method at 70% orientation accuracy. Rules that also see node degrees can exceed it.

Lean: `accRule_sortRule`, `accRule_le_sRho`, `edgeAcc_sortRule_strict`.

Table 2. B1, heterogeneous noise, raw data; means over 180 data sets per family (30 graphs $\times$ 2 p $\times$ 3 T). OA: order accuracy of rel-SortnRegress (rel) and Var-SortnRegress (var), G1 / G2. ΔOA: G1−G2 for rel, with bootstrap 95% CI. SC: strongly connected.

family	exact s_ρ	rel OA	ΔOA rel	var OA
symER (SC)	.78/.50	.63/.50	+.126 [.110,.142]	.64/.50
ring (SC)	.95/.50	.69/.51	+.180 [.162,.198]	.73/.50
chain	.89/.50	.75/.53	+.217 [.199,.235]	.93/.70
out-tree	.84/.50	.69/.55	+.144 [.125,.162]	.94/.71
in-tree	.79/.66	.74/.67	+.069 [.057,.082]	.95/.72
ER(c=1)	.77/.56	.67/.59	+.085 [.068,.102]	.86/.65
ER(c=2)	.83/.52	.70/.59	+.109 [.093,.127]	.74/.59

3.2. Does the start artefact change benchmark results?

We unrolled 420 summary graphs (30 per family and $p \in \{5, 10\}$; families in Table 2) over $T \in \{3, 5, 10\}$ and drew $n = 1000$ linear-Gaussian trajectories (positive weights, heterogeneous noise) either with the fixed start of Reisach et al. (2026) (G1) or after a latent burn-in of 100 steps (G2; an exact stationary start reproduced G2 to within 0.013). Methods saw the pT columns shuffled, with time labels hidden: rel-, R^2- and Var-SortnRegress (Reisach et al., 2026; 2023), FLOP, PC and a random order. The analysis plan was committed before any algorithm was run (Appendix P). The fixed start raised rel-SortnRegress order accuracy in all five non-strongly-connected families (pooled +0.125, 95% CI $[0.116, 0.134]$; Holm $p \le 1.1 \cdot 10^{-10}$ per family), and the gain tracked the exact population gap of Theorems 2 and 5 across the 42 (family, p, T) cells (Spearman 0.80; slope 0.53, so the estimated counts see about half the population gap). Strong connectivity did not protect short horizons: on rings and symmetrised ER graphs the gain was +0.13 to +0.31 at $T < p$ and still +0.08 to +0.22 in most cells with $T \ge p$, as the exact values $s_\rho(G_T)$ predicted (rings: $s_\rho(G_T) = 1$ for all $T \le p$). The shifts reversed rankings: 11 method pairs swapped significantly on SHD between G1 and G2 (Holm over 1,190 tests; 9 remain when first-slice pairs are excluded), e.g. on chains rel-SortnRegress beat R^2-SortnRegress under G1 and lost under G2. The best method changed in 10 of 42 cells, but FLOP was best on average under both generators. The fixed start also creates var-sortability (Var-SortnRegress +0.20 on raw data, +0.002 after standardisation), which drives the largest swaps (Var-SortnRegress against FLOP and PC on in-trees); this second artefact is outside our theory.

4. Aggregation versus Subsampling

For a stable linear SVAR with independent innovations, nodes with disjoint summary ancestor sets are independent at all lags, and so are any per-node, time-invariant transforms, including block means and subsamples; otherwise the lag-0 covariance is nonzero for almost every parameter (Proposition 13, Appendix C). So the population relatives of contemporaneous values are rel_S for every observation

scheme, and the population relatives sort is the stationary value of Theorem 5 (on non-self-loop edges), not the finite-G_T value. Var- and R^2-sortability behave differently (Theorem 14): block means converge, at the textbook Fejér rate for averages of a stationary series (Brockwell & Davis, 1991, Ch. 7), to the long-run structural model of Gong et al. (2017), whereas subsamples keep the snapshot covariance. As a consequence, for large block size, block-mean R^2-sortability is 0 on out-stars and 1 on in-stars with at least two arms for all weights; on out-stars subsampled data are R^2-sorted correctly on average (0.84) while block means drift to 0. We claim only these sortability consequences.

5. Finite-Sample Theory

We now ask whether the population relatives can be estimated. X is a zero-mean stationary Gaussian process with absolutely summable auto- and cross-covariances $\gamma_{ab}(h)$, standardised so that $\gamma_{aa}(0) = 1$, observed at n times. Two nodes are declared relatives if the Fisher-z test of the sample correlation $\hat{r}_{ab}$ rejects at level α with Bonferroni over $m = d(d-1)/2$ pairs, i.e. $|\hat{r}_{ab}| > \tanh(z_c/\sqrt{n'-3})$, $z_c = \Phi^{-1}(1-\alpha/2m)$; $\hat{\rho}(v) = 1+\#$declared partners. The *nominal* test uses $n' = n$. Let $B_{ab} = \sum_h |\gamma_{aa}(h)\gamma_{bb}(h) + \gamma_{ab}(h)\gamma_{ab}(-h)|$ (for an independent pair, the Bartlett sum $\sum_h \rho_a(h)\rho_b(h)$).

F1: recovery and effective sample size. That the nominal-n test has the wrong size under autocorrelation is classical: effective degrees of freedom go back to Bartlett (1935), with spatial and multivariate versions in Clifford et al. (1989) and Roy (1989), and the variance with exactly our summand B_{ab} is in Afyouni et al. (2019). We add a non-asymptotic exact-recovery bound for the relatives set (Theorem 19, Appendix D): if the thresholds separate the population correlations by more than explicit deviations, the declared set and hence $\hat{s} = s_\rho^{\mathrm{pop}}$ are recovered with failure probability decaying as $\exp(-n\min(\tau, \rho_{\min}-\tau)^2/(2(\sqrt{B}+\rho_{\max}\sqrt{D_\gamma})^2))$, so n/B acts as an effective sample size. For i.i.d. snapshots $B = 1+\rho^2$; under strong persistence B can be in the tens.

Proposition 8 (size distortion of the nominal test)**.** *In this setting, a non-relative pair is declared by the nominal test with probability* $\to 2\bar{\Phi}(z_c/\sqrt{W_{ab}})$, $W_{ab} = \sum_h \gamma_{aa}(h)\gamma_{bb}(h)$ *(= B_{ab} when all products are nonnegative), above the nominal α/m when $W_{ab} > 1$. With $n_{\mathrm{eff}} = n(1-\rho^2)^2/W_{ab}$ (W the delta-method Bartlett variance) the false-positive rate tends to α/m.*

Lean: not machine-checked (uses the multivariate Breuer–Major theorem).

The joint asymptotic normality of the two sample variances and the sample cross-covariance is not an assumption: it follows from the Breuer–Major theorem for stationary Gaussian vector sequences (Arcones, 1994), in the form stated by Nourdin et al. (2011, Thm. 1.1), applied through the Cramér–Wold device to linear combinations of the three products, which have Hermite rank 2; its condition $\sum_h |\gamma_{ab}(h)|^2 < \infty$ for all a, b follows from absolute summability (Appendix D).

F2: saturation by a latent driver. Let $X_v = Y_v + b_v C$ with C a stationary unit-variance process independent of the VAR Y, and $\ell_v = b_v^2/\mathrm{Var}X_v$ the driver's share of variance.

Proposition 9 (saturation)**.** *For summary non-relatives, $\rho_{uv} = \mathrm{sgn}(b_u b_v)\sqrt{\ell_u \ell_v} =: \kappa_{uv}$. If all $b_v \neq 0$, every pair is a population relative and the population rel-sortability is exactly $1/2$; $P(\hat{s}_n \neq \frac{1}{2}) \leq (d^2+3d)e^{-cn}$ with $c \approx \kappa_{\min}^2/(18(\sqrt{B}+\sqrt{D_\gamma})^2)$. Asymptotically, a pair is detected with probability $> 1/2$ once $n > n_{1/2} = 3 + (z_c/\operatorname{atanh}\kappa)^2 \approx z_c^2/\ell^2$ (nominal test; for the effective test, the same bound on n_{eff}).*

Lean: `prop21_corr_std`, `sOf_saturated`, `abs_sOf_sub_half_le`, location of $n_{1/2}$: `ncrit_eq_iff`; the probability statements are not machine-checked.

$n_{1/2}$ does not depend on autocorrelation, which only sets the width of the transition. A driver with share $\ell = 0.1$ saturates at $n \approx 10^3$, and one with $\ell = 0.64$ at $n \approx 22$ (both for $z_c = 3.29$). A relatives count on a long record can recover the summary relatives only while $\max_{\text{non-rel}} |\kappa| < \tau_n < \min_{\text{rel}} |\rho|$, a window that closes at n of order $n_{1/2}$.

F3: conditioning on a proxy (Proposition 20, Appendix D). For a proxy $Z = \sqrt{r}\,C + \sqrt{1-r}\,e$ with reliability r, the partial correlation of summary non-relatives given Z is strictly decreasing in r and vanishes only at $r = 1$, so spurious conditional relatives appear for every $r < 1$ once n is large; fewer than half of the non-relative pairs are declared only if $r > r_{\mathrm{crit}} = 1 - \frac{\tau'(1-\ell)}{\ell(1-\tau')}$, $\tau' = \tanh(z_c/\sqrt{n'-4})$ (equal shares; $r_{\mathrm{crit}} = 0.95$ for $\ell = 0.64$, $\tau' = 0.082$). A driver that reaches nodes through different filters leaves a residual even at $r = 1$. The reliability is identified by $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$, the identity behind the three-indicator rule for one-factor models (Bollen, 1989, Ch. 7). Formalisation found four missing nondegeneracy hypotheses in our earlier statements (e.g. $b_u b_v \neq 0$; Appendix F). The non-asymptotic bounds are loose (factors 3–13 in n, Appendix E); the asymptotic statements matched every simulation to within 0.03 and are used below.

6. Machine-Checked Proofs

The formalisation (Lean 4, de Moura & Ullrich, 2021, with Mathlib, The mathlib Community, 2020; 3,963 lines in 16 files) certifies the exact and stationary results of Section 3 (Theorems 1, 2 and 5, Corollaries 3 and 6, Proposition 4, the loop-free exact edge value and the accuracy reading), the algebra of the real-data estimand $\tilde{s}$ (Appendix H), and the

deterministic algebra of F1–F3. It does *not* certify the ER limits, the loop-free limit and periodicity, the lag-weighted extension, or any probability bound or limit theorem. The axiom report lists 148 declarations, each depending only on `propext`, `Classical.choice` and `Quot.sound`, and the sources contain no `sorry`, `admit` or `axiom`: 29 headline theorems (Lean counterparts of numbered statements), 43 supporting graph lemmas, 65 algebraic identities and inequalities of the aggregation and finite-sample parts (many elementary), and 11 examples. Relatives, s_ρ, G_T and the stationary windows are defined from scratch following Reisach et al. (2026). Appendix F maps every statement to its Lean names and status.

7. Real-Data Evidence

Protocol and its limits. We ran four studies (P1–P9, then E1/E2), each with an analysis plan (predictions, thresholds, estimator, seeds, kill rules) committed to version control before any dependence statistic of the study's data was computed in this project (timeline in Appendix I). Three qualifications matter. First, the plans, analyses and results were produced by AI agents within about six hours of one day, and commit timestamps are set by the committing client; no third-party registry was used, so we call them *time-stamped analysis plans*, not pre-registrations. Second, an earlier exploratory project in the same repository (one day before plan 1) had already analysed CausalRivers with the same regions, data loaders and subgraph sampler, at resolutions including ours, and found unrelated stations about as correlated as river siblings (VAR(1) residual correlation 0.51 against 0.51; Appendix I); the CausalRivers predictions were therefore not made blind to that dependence structure, and our F2 account of their failure is explanatory, not predictive. Third, the graph refinement used by E1 was motivated by study-3 observations on the same device.

Estimand. For an observed node set O of a known summary graph G_{S}, the predicted value $\tilde{s}$ is the edge average, over the induced non-self-loop edges, of the scores of the counts $|\mathrm{rel}_S(v) \cap O|$: population lag-0 relatives among the observed nodes, with unobserved nodes acting as latent common ancestors (Definition 21). It is what a perfect lag-0 dependence oracle would give under Proposition 13. It is not the rel-sortability of any unrolled graph. Every scored edge has value $1/2$ or 1, and $\tilde{s} > 1/2$ iff some induced edge (i, j) has an observed node in $\mathrm{rel}_S(j) \setminus \mathrm{rel}_S(i)$; this forces an up-edge $(i, j) \in E_\uparrow$ among the induced edges, but not conversely, because the extra relatives of j may all be unobserved (Proposition 22, machine-checked). The estimator is the relatives count of Section 5 at $\alpha = 0.01$ with Bonferroni; orientation accuracy is the mean edge score of $\hat{\rho}$. CIs are cluster bootstraps (2,000 resamples). Lohse & Wahl (2025) measured var- and R^2-sortability on the same

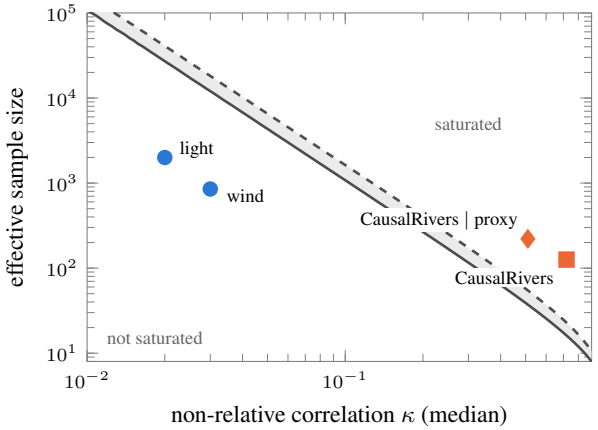


Figure 3. Saturation regime of Proposition 9. Curves: onset $n_{1/2}$ for $z_c = 3.29$ ($d = 5$, solid) and 4.04 ($d = 20$, dashed); above them a non-relative pair is detected with probability $> 1/2$. Points: medians from the real-data studies (CausalRivers unconditioned and given the regional proxy, κ replaced by $|\pi|$; Causal Chambers wind and light tunnel). Values are in Section 7 and Appendix G.

families; we know of no earlier test of rel-sortability on real data.

CausalRivers: a failed test, explained post hoc. River networks are in-forests, so Corollary 6 predicts $\tilde{s} > 1/2$ for subgraphs with confluences (mean 0.82). The estimated relatives sort carried no orientation information (P2, P3 fail; Table 3) because 98% of non-relative station pairs tested dependent. A post hoc reading through F2: non-relative correlations $\kappa \approx 0.64$–0.74 give $n_{1/2} \approx 20$–27, while the records have $n = 377$–$3{,}652$ steps, so the study sat far inside the saturated regime (Figure 3), where Proposition 9 predicts $\hat{s} = 1/2$. Regional precipitation acts as a latent driver; adding it to the summary graph leaves one source per component, and Corollary 6 then predicts exactly $1/2$. The second study's plan fixed the F3 remedy (condition on a leave-out regional mean). It also failed (P4: 0.74). F3 is consistent with this failure: the triad estimate of reliability is $\hat{r} \approx 0.62$, below the $r_{\mathrm{crit}} \approx 0.87$ required at the observed n_{eff}, and the plug-in prediction of the post-conditioning dependent fraction is 0.743 (0.736) against an observed 0.741 (0.735) at 12 h (24 h); this check is in-sample, because $\hat{r}$ and $\hat{\ell}$ come from the tested correlations. Post hoc, 10 principal components still leave 49% of non-relative pairs dependent, consistent with a driver of rank > 1.

Causal Chambers: designed independent sources. The third study chose data in which Corollary 6's non-trivial case holds by design: in the wind-tunnel random-walk runs (Gamella et al., 2025) five actuators are independent experimenter-generated random walks, so they are separate source SCCs; 25 of 26 edges are predicted to be strictly sorted. Saturation is absent (P7, 0.039) and the relatives count orients confluence edges above chance (P8, 0.656). The graded agreement P9 missed its threshold ($0.267 < 0.3$;

Table 3. Tests fixed in the time-stamped analysis plans. Pass/fail is by the point rule fixed in the plan. Brackets: 95% cluster-bootstrap CI. "non-rel. dep.": fraction of graph non-relative pairs declared dependent. "Acc.": orientation accuracy on edges predicted to be strictly sorted.

study (units)	test (threshold)	result (12 h / 24 h)	verdict
CausalRivers, marginal (120 subgraphs)	P1: confluence subgraphs have $\tilde{s} > 1/2$	112 of 114	fails as worded[a]
	P2: Acc. > 0.5, CI excl. 0.5	$0.501\,[0.492, 0.509]$ / $0.498\,[0.489, 0.507]$	**fail**
	P3: Spearman$(\tilde{s}, \text{acc.}) \geq 0.3$	-0.018 / 0.018	**fail**
	(non-rel. dep., nominal test)	0.984 / 0.978	kill rule triggered
CausalRivers, conditioned on regional proxy (120 subgraphs)	P4: non-rel. dep. < 0.5 given proxy	0.741 / 0.735	**fail**
	P5: Acc. > 0.5, CI excl. 0.5	$0.539\,[0.513, 0.565]$ / $0.540\,[0.514, 0.567]$	pass, not interp.[b]
	P6: Spearman ≥ 0.3	-0.113 / -0.121	**fail**
Causal Chambers, wind tunnel (15 runs, primary)	P7: non-rel. dep. < 0.5	$0.039\,[0.031, 0.047]$	pass
	P8: Acc. > 0.5, CI excl. 0.5	$0.656\,[0.637, 0.675]$	pass
	P9: Spearman ≥ 0.3 (300 subgraphs)	$0.267\,[0.163, 0.368]$	**fail**
light tunnel (10 segments, replication)	R7 / R8 / R9 (same thresholds)	0.000 / 1.000 / $0.779\,[0.696, 0.864]$	pass (replication)[c]
E1, light tunnel (10 units, G_{lin})	E1a: non-rel. dep. < 0.5; det. ≥ 0.8	0.000; $0.969\,[0.945, 0.992]$	pass
	E1b: $\Delta \leq -0.10$, Holm $p \leq 0.05$, all 8	$\Delta = -0.502$ to -0.533 (10/10)	pass (8/8)[d]
	E1c: as E1b, detected edges	$\Delta = -0.500$	pass (8/8)
E2, wind regimes (64 units; D=1, lags L=30 vs. 0)	Q7: non-rel. dep. < 0.5	$0.003\,[0.001, 0.005]$	pass
	Q8: Acc. > 0.5, CI excl. 0.5	$0.553\,[0.527, 0.579]$	pass
	Q9: Spearman ≥ 0.3 (1,280 subgr.)	$0.243\,[0.193, 0.294]$	**fail**
	Q10: detection gain, CI lower bound > 0	$+0.069\,[0.060, 0.079]$	pass
	E2 success (Q7–Q10 all)		**not met**

[a]The two exceptions are split-and-rejoin channels with a single source, for which the theorem itself predicts $1/2$. [b]Kill rule (P4 failed at both resolutions): P5 is reported, not interpreted. [c]Fixed in the plan as not counting towards a positive demonstration (i.i.d. inputs, weak time-series aspect). [d]G_{lin} is the primary graph fixed in plan 4. It omits six real causal edges that, under a symmetry lemma and an assumption of ideal polariser sensors, carry no linear dependence. On the published graph the theorem still beats all eight overall, but not on its strictly ordered edges, and on G_{lin} the prediction equals a weak-component-size rule (Section 7, Appendix J).

Table 4. Out-of-sample F-theory check on the Causal Chambers: predictions from the first half of each unit, observed rates on the second half. Accuracy is predicted by Monte Carlo of independent pair decisions.

	wind tunnel		light tunnel	
	pred.	obs.	pred.	obs.
non-relatives declared (F2)	0.031	0.028	0.006	0.000
relatives detected (F1)	0.360	0.343	0.679	0.681
orientation accuracy	0.630	0.620	0.999	1.000

CI contains 0.3), so by the plan's own definition the study is *not* a positive real-data demonstration. The reason is F1, not a driver: 16 of the 26 ground-truth edges are essentially undetectable at lag 0 at $n \approx 10^3$, which compresses the estimated counts. The light-tunnel replication with i.i.d. inputs passes all three criteria. F-theory predictions computed from the first half of each unit and evaluated on the second half are accurate to within 0.02 (Table 4); F2 predicts no saturation ($\hat{\kappa} \approx 0.03$ and 0.02 give $n_{1/2} \approx 15{,}600$ and $33{,}800$, far above $n_{\text{eff}} \approx 430$–$1{,}000$). The wind-tunnel graph is bipartite, so the prediction coincides with "sensors have more relatives than actuators"; all units of a chamber share one device.

E1: the theorem against naive predictors (details in Appendix J). Plan 4 tested the edge values of Corollary 6 against eight naive predictors (sensor indicator; ancestor count, in-degree and depth on the full and induced graph; "every edge 1") on the light-tunnel series `citris_1` (10 units), with absolute loss on disagreeing edge instances, an exact sign-flip test and Holm correction. On the primary graph G_{lin} it passed 8/8 ($\Delta = -0.502$ to -0.533, 10/10 units, Holm $p = 0.0078$, the smallest attainable; exact match 99.2% of 1,662 instances against 11–20%). This result is weaker than it looks. (i) G_{lin} removes six real causal edges that, under a symmetry lemma assuming ideal polariser sensors, carry no linear dependence; on the published graph the sensor indicator is significantly better on the 442 edges predicted strict, and a constant-$1/2$ rule (post hoc) matches more instances than the theorem (81.5% against 71.4%). (ii) Each weak component of G_{lin} has a single source, so every node's relatives are its whole weak component; on all 1,662 instances the theorem's prediction therefore coincides with the trivial rule "tie iff same weak component, otherwise compare the observed sizes of the two components", which was not among the competitors. (iii) Most of the separation comes from ties (1,365 of the predictions are $1/2$, which no competitor can output). (iv) The data are 10 units of one series on one device; on 64 wind-tunnel regimes the theorem again beats all eight, but a constant-$1/2$ rule has lower absolute error (0.311 against

0.363). E1 thus shows that the component structure of G_{lin} predicts the observed ties and orderings; it does not discriminate the theory from that simpler rule.

E2: lagged relatives (Appendix K). A Bonferroni max-over-lags variant of classical cross-correlation independence tests (Haugh, 1976) ($|h| \leq 30$ on first differences, against lag 0) on 64 held-out wind-tunnel regimes raised detection from 0.328 to 0.398 without more false positives, but failed the graded criterion ($0.243 < 0.3$), so E2 does not succeed. The lag-0 estimator failed that criterion again on these regimes, so P9's verdict stands.

8. Discussion and Limitations

For benchmark design. The finite time-unrolled generator is rel-sortable above $1/2$ for every summary graph that is not a union of strongly connected components (Corollary 3), and part of the excess comes only from the start at $t = 1$. Benchmarks should use strongly connected summary graphs or a burn-in of at least Δ slices ($\Delta\tau_{\max}$ with lags up to $\tau_{\max}$; Δ cannot be lowered in general), after which the value is the graph property of Theorem 5; strong connectivity alone does not suffice at the short horizons proposed for benchmarks (Section 3.2). Columns should also be standardised, since the fixed start inflates var-sortability as well. A latent burn-in makes its nodes latent common causes, so a ground truth that assumes causal sufficiency must be stated for the window graph with this in mind; an algorithm that sees time stamps orients every cross-time edge for free, so the artefact matters for methods that do not use time order. Random summary graphs are sortable only near criticality.

For real data. Population rel-sortability above $1/2$ does not imply that a relatives count can be estimated. Two conditions are needed: no latent common driver with share ℓ such that $n \gtrsim z_c^2/\ell^2$ (or a near-perfect, rank-one proxy), and relatives detectable at lag 0 at the available n_{eff}. Diagnostic: compare n_{eff} with $3+(z_c/\operatorname{atanh}\hat{\kappa})^2$, where $\hat{\kappa}$ is the typical correlation of pairs believed unrelated; if n_{eff} is far above it, the count is saturated whatever the graph.

Limitations. The closed forms assume self-loops at every node and unit lags (Appendices L and M). The ER limits are paper proofs not yet checked by a human probabilist; they are iterated limits ($T \to \infty$ at fixed p, then $p \to \infty$; the stationary value needs no T-limit), with no rates and no result for fixed T. Section 4 covers linear SVARs only. The finite-sample bounds are loose; Proposition 13 and the probabilistic parts of F1–F3 are proved on paper only, for Gaussian processes. The analysis plans are internally time-stamped, agent-executed on one day, and the CausalRivers plan was not blind to an earlier analysis of the same data. The positive real-data evidence comes from one device per chamber; E1 coincides with a weak-component-size rule on its primary graph and is carried mainly by ties. The graded criteria P2, P3, P4, P6, P9 and Q9 failed. B1 uses one linear-Gaussian parametrisation with positive weights and $n = 1000$; its ranking reversals involve sort-based baselines, not the best method, and the largest ones come from var-sortability rather than rel-sortability.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Statement on the Use of AI Tools (draft for the authors)

Most of this project was carried out with AI coding agents (Claude-based) working under the authors' direction: the paper proofs (including Appendix N.1), all Lean code, all experiment and analysis code, the analysis plans, the simulated pre-submission review that motivated this revision, and the manuscript text were AI-generated. What is machine-checked: the 148 Lean declarations of Section 6 (Lean kernel, standard axioms only), which cover Section 3's exact and stationary results, the estimand algebra and the deterministic algebra of F1–F3. What is checked only by computation: the enumeration of Appendices B and L, the ER numerics and the simulations of Appendix E. What no human has yet read line by line: the paper-only proofs, namely Appendix N.1, Proposition 13, Theorem 14, the probabilistic parts of Theorem 19 and Propositions 8, 9 and 20, the loop-free limit results of Appendix L and the lag-weighted extension (Appendix M). The authors take responsibility for all content.

Artefact Statement

Lean code, verification scripts, analysis scripts, analysis plans and run logs are provided as supplementary material (anonymised). Raw data are public (CausalRivers; Causal Chambers, CC BY 4.0) and are not redistributed.

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Appendix overview

Appendix A gives full proofs of Section 3, Appendix B the exhaustive enumeration check, Appendix C the proofs of Section 4, Appendix D the proofs of Section 5, Appendix E the simulation checks of F1–F3, Appendix F the map from statements to Lean names, Appendix G details of the real-data studies, Appendix H the real-data estimand $\tilde{s}$, Appendix I the analysis-plan timeline, prior exposure and deviations, Appendices J and K the experiments E1 and E2 in full, Appendices L to O the loop-free graphs, lagged unrollings, the random summary graphs (with the full proof of Proposition 7 in Appendix N.1) and the accuracy interpretation.

Status labels. *Machine-checked*: a Lean theorem with no `sorry` and only the standard axioms. *Proved*: a complete paper proof given here. *Asymptotic*: a limit statement proved here from the multivariate Breuer–Major theorem (Arcones, 1994; Nourdin et al., 2011) (cited, not re-proved). *Sketched*: a derivation with one analytic step left out, stated as such (used only for statements outside the main claims: the windowed loop-free stationary value, the layered-DAG stationary value and the constants of the lagged recovery bound). *Enumerated*: checked by exact computation on many instances, not proved.

A. Proofs for Section 3

Definition of s_ρ, verbatim. Reisach et al. (2026, Sec. 2.2) (arXiv v1) write: "For a graph $G = (V, E)$ with $|E| \geq 1$ we define sortability by the number of relatives, or rel-sortability for short, as $s_\rho(G) := \frac{\sum_{(x,y)\in E} s_\rho(x,y)}{|E|}$ with $s_\rho(x, y) := 0$ if $\rho(x) > \rho(y)$, $\frac{1}{2}$ if $\rho(x) = \rho(y)$, 1 if $\rho(x) < \rho(y)$, for $(x, y) \in E$." (their Eq. (4); the cases are typeset as a brace there), and add: "Note that our definition differs from the sortabilities in these works in that we count edges instead of paths." Our s_ρ is this edge average; in G_T the edges are the copies $(i(t), j(t{+}1))$.

Throughout, $G_S = (V, E_S)$ has a self-loop at every node, $p = |V| \geq 2$, and $d(u, v)$ is as in Section 2. We use two facts of Reisach et al. (2026), proved there for DAGs: (Thm. 1) if $x \in \mathrm{anc}(y)$ then $\mathrm{rel}(x) \subseteq \mathrm{rel}(y)$; (Lemma 1, eq. (10)) if $x \in \mathrm{anc}(y)$ then $\mathrm{rel}(x) \subsetneq \mathrm{rel}(y)$ iff $\mathrm{roots}(y) \setminus \mathrm{roots}(x) \neq \emptyset$, where $\mathrm{roots}(y) = \{x \in \mathrm{anc}(y) : \mathrm{anc}(x) = \{x\}\}$. Both are also machine-checked independently (`relSet_subset_of_edge`, and the direct characterisation `isRel_unrolled_iff`).

Lemma 10 (ancestors in G_T)**.** *$u(s) \in \mathrm{anc}(v(t))$ iff $s \leq t$ and $d(u, v) \leq t - s$.*

Lean: `unrolled_reach_iff`, `unrolled_reach_iff_dist`.

Proof. Every edge of G_T increases time by one, so a directed path from $u(s)$ to $v(t)$ has $t - s$ edges, and its projection to V is a walk of length $t - s$ from u to v in G_S (self-loops allowed as steps). Such a walk exists iff $d(u, v) \leq t - s$: a walk contains a path, so its length is at least $d(u, v)$; conversely, pad a shortest path with $t - s - d(u, v)$ self-loop steps at u. Every such walk lifts to a directed path in G_T. □

Lemma 11 (roots)**.** $\mathrm{roots}(v(t)) = \{u(1) : d(u, v) \leq t - 1\}$, *and* $\mathrm{desc}(u(1)) = \{w(s) : d(u, w) \leq s - 1\}$.

Lean: `roots_unrolled_iff`, `isRoot_unrolled_iff`, `desc_root_unrolled_iff`.

Proof. A node $u(s)$ with $s \geq 2$ has the parent $u(s - 1)$ through the self-loop, so it is not a root; $u(1)$ has no parent. Apply Lemma 10 with $s = 1$, in both directions. □

Proof of Theorem 1. $i(t)$ is a parent of $j(t{+}1)$, so $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$ and $s_\rho(e) \in \{1/2, 1\}$, with value 1 iff the inclusion is strict, iff $\mathrm{roots}(j(t{+}1)) \setminus \mathrm{roots}(i(t)) \neq \emptyset$. By Lemma 11 this set is $\{u(1) : d(u, j) \leq t,\ d(u, i) > t - 1\}$, and since d is integer-valued or ∞, $d(u, i) > t - 1$ iff $d(u, i) \geq t$. □

Proof of Proposition 4 (and the general count). Since every ancestor of y descends from a root of y that is itself an ancestor, $\mathrm{rel}(y) = \mathrm{desc}(\mathrm{roots}(y))$. By Lemma 11, $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d(u, v) \leq t - 1$ and $d(u, w) \leq s - 1$, i.e. iff $s \geq 1 + m_t(v, w)$ with $m_t(v, w) = \min\{d(u, w) : d(u, v) \leq t - 1\}$. Counting $s \in \{1, \ldots, T\}$ gives $\rho(v(t)) = \sum_w \max(T - m_t(v, w), 0)$ (`rho_unrolled_eq`). For $t \geq p$ every finite $d(u, v)$ is at most $p - 1 \leq t - 1$, so the admissible u are $\mathrm{anc}_S(v)$ and $m_t(v, w) = d(\mathrm{anc}_S(v), w)$, finite iff $w \in \mathrm{rel}_S(v)$ and then at most $p - 1 < t \leq T$. (The Lean statement `rho_unrolled_large_t` needs only $t \geq p$; an earlier hypothesis $T \geq p$ is redundant.) □

Proof of Theorem 2. Crossing edge $(i, j) \in E_\times$: j does not reach i, so $d(j, i) = \infty$, and $u = j$ gives $d(j, j) = 0 \leq t \leq \infty$ for every t: all $T - 1$ copies score 1. Edge in $E_\circ$: i and j reach each other (or $i = j$), so for every u, $d(u, i) < \infty$ iff

$d(u,j) < \infty$. If the condition of Theorem 1 held with $d(u,i) = \infty$ then $d(u,j) = \infty > t$, a contradiction; hence it forces $t \le d(u,i) \le p-1$, and at most $N_T(i,j) \le \min(p-1, T-1)$ copies score 1. Summing over the $(T-1)|E_{\mathrm{S}}|$ edges,

$$s_\rho(G_T) = \tfrac{1}{2} + \frac{1}{2(T-1)|E_{\mathrm{S}}|}\Big[(T-1)|E_\times| + \sum_{(i,j)\in E_\circ} N_T(i,j)\Big],$$

which gives both bounds; let $T \to \infty$. □

Proof of Corollary 3. $E_\times = \emptyset$ iff no edge joins two SCCs. A weakly connected component made of two or more SCCs is connected by undirected edges, one of which joins two different SCCs; conversely a strongly connected component has no crossing edge. The uniform bound is Theorem 2 with $|E_\times| \ge 1$. For the DAG case, every SCC is a singleton, $E_\times$ consists of the m non-loop edges, and $|E_{\mathrm{S}}| = m + p$. □

Proof of Theorem 5. In G_{B+T}, by Lemma 11 shifted by B, $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d(u,v) \le t-1$ and $d(u,w) \le s-1$. For window nodes $t-1, s-1 \ge B \ge p-1$, and every finite distance is at most $p-1$, so the condition is $u \in \mathrm{anc}_S(v) \cap \mathrm{anc}_S(w)$, independent of s,t. Hence $\mathrm{rel}_W(v(t)) = \{w(s) \in W : w \in \mathrm{rel}_S(v)\}$, of size $T|\mathrm{rel}_S(v)|$. On $V \times \mathbb{Z}$, Lemma 10 holds verbatim and the common ancestor can be taken arbitrarily early, giving the same set. Each copy of (i,j) therefore compares $|\mathrm{rel}_S(i)|$ with $|\mathrm{rel}_S(j)|$; as $i \in \mathrm{anc}_S(j)$, $\mathrm{rel}_S(i) \subseteq \mathrm{rel}_S(j)$, so the score is 1 iff $\mathrm{rel}_S(i) \ne \mathrm{rel}_S(j)$, and $1/2$ for self-loops. Source SCCs: $\mathrm{rel}_S(j) \setminus \mathrm{rel}_S(i) \ne \emptyset$ iff $\mathrm{anc}_S(j) \not\subseteq \mathrm{rel}_S(i)$ (if $z \in \mathrm{rel}_S(i)$ then $\mathrm{desc}_S(z) \subseteq \mathrm{rel}_S(i)$). Every node of $\mathrm{anc}_S(j)$ is reached from some $\sigma \in \mathrm{Src}(j)$, and $\mathrm{desc}_S(\sigma) \subseteq \mathrm{rel}_S(i)$ whenever $\sigma \subseteq \mathrm{rel}_S(i)$. A source SCC meets $\mathrm{rel}_S(i)$ iff it reaches i (its only ancestors are its own nodes), and then it is contained in $\mathrm{rel}_S(i)$. So the condition is $\mathrm{Src}(j) \not\subseteq \mathrm{Src}(i)$, which, as $\mathrm{Src}(i) \subseteq \mathrm{Src}(j)$, is $\mathrm{Src}(i) \ne \mathrm{Src}(j)$. □

Proof of Corollary 6. If each weak component C has a unique source SCC σ_C, then $\mathrm{Src}(v) = \{\sigma_C\}$ on C and $E_\uparrow = \emptyset$. Conversely, if $E_\uparrow = \emptyset$, Src is constant along edges and hence on each weak component; a node v of a source SCC σ has $\mathrm{Src}(v) = \{\sigma\}$, so there is only one. Out-forests (at most one parent besides itself; acyclicity is not needed, `stat_half_of_outForest`) and strongly connected graphs (`stat_half_of_strong`) have one source SCC per component. In an in-forest (acyclic apart from self-loops, out-degree at most one), if $u \in \mathrm{anc}(v)$ then $\mathrm{desc}(u)$ is the unique downstream path through v, so $\mathrm{rel}(v) = \mathrm{anc}(v) \cup \mathrm{desc}(v)$ (`inForest_isRel_iff`); the source SCCs are the headwaters, and $\mathrm{Src}(j) = \mathrm{Src}(i) \cup \bigcup_k \mathrm{Src}(k)$ over the other parents k of j, with disjoint headwater sets on different branches, so $\mathrm{Src}(j) \ne \mathrm{Src}(i)$ iff j is a confluence (`inForest_up_iff`). Out-degree at most one alone is not enough: for $a \rightleftarrows b$ with $c \to a$, a has two parents but the only source SCC is $\{c\}$. □

Remark 12 (self-loops). Without self-loops the lemmas fail: for the single edge $0 \to 1$ without loops, $s_\rho(G_T) = 1/2$ for every $T \ge 2$ (`edgeNoLoops_sRho`); with loops the limit is $2/3$ (`edgeLoops_limit`).

B. Exhaustive Enumeration Check

Before formalisation, every statement of Section 3 was checked by brute force (bitset relatives with the definitions of Reisach et al., 2026) on all $4 + 64 + 4096$ summary graphs with $p = 2, 3, 4$ (all $T \le 9$) and on 10,000 random graphs ($p = 2$–7; Erdős–Rényi, DAG, in-forest, out-forest, chain with back edge and block-SCC families; three values of $T \le 14$ each, fixed seed). In total 46,724 (graph, T) cases and 1,895,609 edges were checked with zero failures of Theorem 1, Proposition 4, the bounds of Theorem 2, Theorem 5 (with $B = p-1$ and $B = 2p+1$) and the source-SCC characterisation. The conjecture fixed in plan 1 ("an edge is strictly sorted in the limit iff $|\mathrm{rel}_S(j)| > |\mathrm{rel}_S(i)|$") disagreed with the exact G_T limit in 9,603 random graphs and never with the stationary value; the conjecture is Theorem 5, not the limit of G_T.

C. Proofs for Section 4

Assumptions: $X(t) = \sum_{\tau=1}^{L} A_\tau X(t-\tau) + \varepsilon(t)$, $t \in \mathbb{Z}$, i.i.d. innovations with diagonal positive-definite covariance Σ, stable (all roots of $\det(I - \sum_\tau A_\tau z^\tau)$ outside the unit disc), stationary solution; the summary graph has $i \to j$ iff $(A_\tau)_{ji} \ne 0$ for some τ. $\Gamma(h)$ is the autocovariance, $A(1) = \sum_\tau A_\tau$, and for a block size D the block means are $\bar{Y}_s = \frac{1}{D}\sum_{u<D} X(sD+u)$ and the subsamples $Z_s = X(sD)$.

Proposition 13 (relatives survive any per-node filter)**.** *If $\mathrm{anc}_S(i) \cap \mathrm{anc}_S(j) = \emptyset$ then X_i and X_j are independent at all lags, and so are any per-node, time-invariant (possibly nonlinear) transforms of them, including $\bar{Y}$ and Z for every D. If the ancestor sets intersect, $\Gamma(0)_{ij}$, $\mathrm{Cov}(\bar{Y})_{ij}$ and the long-run covariance are nonzero for Lebesgue-almost every parameter.*

Lean: not machine-checked (needs the SVAR moving-average representation and real-analytic genericity).

Theorem 14 (block means vs subsampling). *(a)* $\|D\,\mathrm{Cov}(\bar{Y}) - S_{LR}\| \le \frac{1}{D}\sum_h |h|\|\Gamma(h)\|$, *with* $S_{LR} = (I - A(1))^{-1}\Sigma(I - A(1))^{-\top}$. *(b) If* G_{S} *is acyclic apart from self-loops,* S_{LR} *is the covariance of a static SEM (this long-run SEM is due to Gong et al., 2017, Eqs. (5), (10)) on* G_{S} *with weights* $A(1)_{ji}/(1-\lambda_j)$ *and noise variances* $\sigma_j^2/(1-\lambda_j)^2$, $\lambda_j = A(1)_{jj} < 1$. *(c)* $\mathrm{Cov}(Z_s) = \Gamma(0)$ *for every* D. *(d) Hence, for large* D, *block-mean* R^2*-sortability is* 0 *on out-stars and* 1 *on in-stars with* ≥ 2 *arms, for all weights; under subsampling every contemporaneous criterion keeps its* $\Gamma(0)$ *value.*

Lean: (a) counting identity `fejer_identity`*; (b)* `longrun_factor`, `selfloop_sum_lt_one`*; the probabilistic steps, the rate and (c)–(d) are not machine-checked.*

Part (a) is the standard Fejér/Cesàro computation for the variance of the mean of a stationary series (Brockwell & Davis, 1991, Ch. 7), written for block means of a vector process; (b) is due to Gong et al. (2017); (c) is stationarity. We claim only the sortability consequences (d).

Proof of Proposition 13. By the MA(∞) representation $X(t) = \sum_k \Psi_k \varepsilon(t-k)$, and because Ψ_k is supported on the reachability relation, $X_i(t)$ is a measurable function of $\{\varepsilon_k(s) : k \in \mathrm{anc}_S(i)\}$. Disjoint ancestor sets mean independent innovation sets, so the processes X_i and X_j are independent, and so are any per-node transforms. For the generic part, each covariance entry is real-analytic in (A, Σ) on the connected open set of stable parameters with the given support, hence either identically zero or nonzero almost everywhere. With all coefficients positive, including the self-loop at a common ancestor k, all Ψ entries are nonnegative and $\Psi_{ik}(m), \Psi_{jk}(m) > 0$ for all large m (self-loops pad the walks), so $\Gamma(0)_{ij} > 0$; the same holds for $\Gamma(h)$, $h \ge 0$, and therefore for the Fejér sums and S_{LR}. "Generic" does not mean "large": with little autocorrelation, $\Gamma(0)_{ij}$ along a lagged edge can be tiny (down to 10^{-14} in our exact computations; exactly 0 without autocorrelation). □

Proof of Theorem 14. (a) $D^2\mathrm{Cov}(\bar{Y}) = \sum_{u,w<D}\Gamma(u-w) = D\Gamma(0) + \sum_{h=1}^{D-1}(D-h)(\Gamma(h)+\Gamma(-h))$ (the counting identity is `fejer_identity`). The stable VAR has $\|\Gamma(h)\| \le Cr^{|h|}$, $r < 1$, so $\sum_h |h|\|\Gamma(h)\| < \infty$ and $S_{LR} = \sum_h \Gamma(h) = \Psi(1)\Sigma\Psi(1)^\top$ with $\Psi(1) = (I - A(1))^{-1}$. Then $S_{LR} - D\,\mathrm{Cov}(\bar{Y}) = \sum_{|h|\ge D}\Gamma(h) + \sum_{|h|<D}\frac{|h|}{D}\Gamma(h)$, and $1 \le |h|/D$ in the first sum. (b) Write $I - A(1) = (I-\Lambda) - N$ with Λ the diagonal and N the off-diagonal part; then $I - A(1) = (I-\Lambda)(I-\tilde{B})$ with $\tilde{B} = (I-\Lambda)^{-1}N$ strictly triangular in a topological order (`longrun_factor`, `longrun_inv`). Under acyclicity the AR polynomial factorises into univariate $\phi_j(z) = 1 - \sum_\tau (A_\tau)_{jj} z^\tau$; stability, $\phi_j(0) = 1$ and the intermediate value theorem give $\phi_j(1) = 1 - \lambda_j > 0$ (`selfloop_sum_lt_one`). (c) Stationarity. (d) For a linear SEM $X = BX + e$ with noise variances ω, $R_i^2 = 1 - 1/(\Sigma_{ii}(\Sigma^{-1})_{ii})$ and $(\Sigma^{-1})_{ii} = 1/\omega_i + \sum_{k\in\mathrm{ch}(i)} B_{ki}^2/\omega_k$. In an out-star with root 0 and leaves k, $1 - R_0^2 = 1/(1+\sum_k r_k)$ and $1 - R_k^2 = 1/(1+r_k)$ with $r_k = \omega_0 B_{k0}^2/\omega_k > 0$, so with ≥ 2 leaves the root has strictly larger R^2 than every leaf; the in-star is symmetric. Strict inequalities in the limit SEM of (b), plus (a) and continuity of $S \mapsto R^2$ at positive-definite S, give the claim for all large D. Under subsampling every contemporaneous statistic is a function of $\Gamma(0)$. □

The R^2 statement for stars strengthens the three-node analysis in Herman et al. (2025, App. D), whose qualitative message (that R^2 reflects connectivity, with an asymmetry between colliders and confounders) we credit. Numerical observations (exact population moments, not proved): mean block-mean R^2-sortability over 150 in-trees moves from 0.853 ($D = 1$) to 0.834 ($D = 4$–8) and 0.880 ($D = 256$), so it is not monotone in D; on out-stars subsampled data are R^2-sorted at 0.842 on average while block means give 0.546 ($D = 4$), 0.101 ($D = 16$) and 0 in the limit.

D. Proofs for Section 5

Notation. $\hat{C}_{ab} = \frac{1}{n}\sum_t (X_a(t) - \bar{X}_a)(X_b(t) - \bar{X}_b)$, $\hat{r}_{ab} = \hat{C}_{ab}/\sqrt{\hat{C}_{aa}\hat{C}_{bb}}$. Constants: B_{ab} as in the main text; $D_\gamma = \max_a 2\sum_h \gamma_{aa}(h)^2$; $\kappa_1 = \max_a \sum_h |\gamma_{aa}(h)|$; $\kappa_2 = \max_{a\ne b}(\sum_h |\gamma_{aa}(h)| + \sum_h |\gamma_{ab}(h)|)$. For i.i.d. sampling, $B_{ab} = 1 + \rho_{ab}^2$, $D_\gamma = 2$, $\kappa_1 = 1$.

Lemma 15 (Gaussian chaos tail). *Let* $g \sim N(0, I_N)$, A *symmetric,* $S = g^\top A g - \mathrm{tr}A$. *For* $x > 0$, $P(S \ge 2\|A\|_F\sqrt{x} + 2\|A\|_{op}x) \le e^{-x}$, *and the same for* $-S$.

Lean: algebraic steps `phi_le`, `chernoff_logmgf_le`, `chernoff_choice`, `sqrt_two_v_x`*; the probability statement is not machine-checked.*

Proof. Diagonalise; S has the law of $\sum_i \lambda_i(g_i^2-1)$. For $2s\lambda<1$, $\log Ee^{s\lambda(g^2-1)}=\varphi(s\lambda)$ with $\varphi(u)=-\frac{1}{2}\log(1-2u)-u=\sum_{k\ge 2}(2u)^k/(2k)\le u^2/(1-2|u|)$ for $|u|<1/2$. With $c=2\|A\|_{op}$, $v=2\|A\|_F^2$ and $0\le s<1/c$, $\log Ee^{sS}\le vs^2/(2(1-cs))$. Markov's inequality with $t=\sqrt{2vx}+cx$, $q=\sqrt{2x/v}$ and $s=q/(1+cq)$ gives exponent exactly x. This is the Gaussian case of Hsu et al. (2012, Prop. 1), who derive it from Laurent & Massart (2000); the proof here is self-contained. □

Lemma 16 (exact variance). *With $\tilde\gamma_{ab}=\frac{1}{n}\sum_t X_a(t)X_b(t)=g^\top Ag$ for $A=C^{1/2}MC^{1/2}$, $M=\frac{1}{2n}\left(\begin{smallmatrix}0 & I\\ I & 0\end{smallmatrix}\right)$,*

$$2n\|A\|_F^2=\sum_{|h|<n}\Big(1-\tfrac{|h|}{n}\Big)\big[\gamma_{aa}(h)\gamma_{bb}(h)+\gamma_{ab}(h)\gamma_{ab}(-h)\big]\le B_{ab},$$

$\|A\|_{op}\le\kappa_2/(2n)$; on the diagonal $2n\|A\|_F^2\le D_\gamma$ and $\|A\|_{op}\le\kappa_1/n$; and $\bar X_a\sim N(0,\bar\sigma_a^2)$ with $\bar\sigma_a^2\le\kappa_1/n$.

Lean: trace and Toeplitz algebra `quad_M`, `trace_MC`, `lemma12_trace`, `lemma12_fejer`, `lemma12_le_B`, `lemma12_diag_le_D`; the step $\|C^{1/2}MC^{1/2}\|_F^2=\mathrm{tr}(MCMC)$ and the operator-norm bounds are not machine-checked.

Proof. $\|A\|_F^2=\mathrm{tr}(MCMC)$ and, with Toeplitz blocks $(C_{ab})_{st}=\gamma_{ab}(s-t)$, $4n^2\mathrm{tr}(MCMC)=2\sum_{s,t}[\gamma_{aa}(s-t)\gamma_{bb}(s-t)+\gamma_{ab}(s-t)\gamma_{ab}(t-s)]$ (using that γ_{aa}, γ_{bb} are even); count pairs with $s-t=h$. $\|A\|_{op}\le\|C\|_{op}\|M\|_{op}$ and $\|C\|_{op}$ is at most the largest absolute row sum. □

Lemma 17 (deviation event). *Let $\varepsilon_{ab}=\sqrt{2B_{ab}x/n}+\kappa_2 x/n+2\kappa_1 y/n$ and $\varepsilon_d=\sqrt{2D_\gamma y/n}+4\kappa_1 y/n$. With probability at least $1-d(d-1)e^{-x}-4de^{-y}$, $|\hat C_{ab}-\rho_{ab}|\le\varepsilon_{ab}$ for all $a\ne b$ and $|\hat C_{aa}-1|\le\varepsilon_d$ for all a.*

Lean: deterministic steps `centred_cov_eq`, `centred_dev_offdiag`, `centred_dev_diag`, `lemma13_eps`, `lemma13_count`.

Proof. $\hat C_{ab}=\tilde\gamma_{ab}-\bar X_a\bar X_b$. Apply Lemmas 15 and 16 to each off-diagonal and diagonal $\tilde\gamma$, bound $|\bar X_a|\le\sqrt{2\kappa_1 y/n}$ by the Gaussian Chernoff bound, and take a union bound. □

Lemma 18 (correlation perturbation). *If $a',b'\in[1-\varepsilon_d,1+\varepsilon_d]$, $\varepsilon_d<1$ and $|c'-\rho|\le\varepsilon$, then $|c'/\sqrt{a'b'}-\rho|\le(\varepsilon+|\rho|\varepsilon_d)/(1-\varepsilon_d)$.*

Lean: `corr_perturb` (the hypothesis $|\rho|\le 1$ of our earlier statement is not needed).

Proof. $|c'/\sqrt{a'b'}-\rho|\le|c'-\rho|/\sqrt{a'b'}+|\rho||1/\sqrt{a'b'}-1|$, with $1/\sqrt{a'b'}\le 1/(1-\varepsilon_d)$ and $|1/\sqrt{a'b'}-1|\le\varepsilon_d/(1-\varepsilon_d)$. □

Theorem 19 (recovery bound). *For $x,y>0$ let ε_{ab}, $\varepsilon_d<1$ be as in Lemma 17 and $\delta_{ab}=(\varepsilon_{ab}+|\rho_{ab}|\varepsilon_d)/(1-\varepsilon_d)$. If thresholds separate ($\delta_{ab}\le\tau_{ab}$ on non-relatives and $|\rho_{ab}|-\delta_{ab}>\tau_{ab}$ on relatives), the declared set equals the true one, and hence $\hat s=s_\rho^{\mathrm{pop}}$, with probability at least $1-d(d-1)e^{-x}-4de^{-y}$. For a common threshold $\tau<\rho_{\min}$ the failure probability decays as $\exp(-n\min(\tau,\rho_{\min}-\tau)^2/(2(\sqrt{B}+\rho_{\max}\sqrt{D_\gamma})^2))$.*

Lean: deterministic core `thm15_det`, `decide_eq_target`, `corr_perturb`, `sOf_cntDecl_congr`; Chernoff and trace algebra `chernoff_choice`, `phi_le`, `lemma12_trace`, `lemma12_le_B`. The Gaussian tail bound itself is not machine-checked.

The proof combines a Gaussian-chaos tail bound (the Gaussian case of Hsu et al., 2012, Prop. 1; see also Laurent & Massart, 2000), an exact finite-n variance identity for $\hat r$, and a deterministic perturbation lemma.

Proof of Theorem 19. Set $\delta_{ab}=(\varepsilon_{ab}+|\rho_{ab}|\varepsilon_d)/(1-\varepsilon_d)$. On the event of Lemma 17, Lemma 18 gives $|\hat r_{ab}-\rho_{ab}|\le\delta_{ab}$, so non-relatives satisfy $|\hat r|\le\tau$ and relatives $|\hat r|>\tau$ (`decide_eq_target`, `thm15_det`). Equal declared sets give equal counts and hence $\hat s=s_\rho^{\mathrm{pop}}$ (`sOf_cntDecl_congr`). For a VAR observed as a long stationary series, s_ρ^{pop} is the stationary value of Theorem 5 on non-self-loop edges (Proposition 13). Explicit form: with $y=x+\log(4/(d-1))_+$ (`explicit_y`) and a common threshold $\tau<\rho_{\min}$, $P(\hat R\ne R)\le 2d(d-1)\exp\big(-n\min(\tau,\rho_{\min}-\tau)^2/(2(\sqrt{B}+\rho_{\max}\sqrt{D_\gamma})^2)(1+o(1))\big)$. □

The multivariate Breuer–Major theorem. We use it in the form stated by Nourdin et al. (2011, Thm. 1.1), who attribute the vector case to Arcones (1994): let $X=(X_k)_{k\in\mathbb{Z}}$ be a centred stationary Gaussian process in $\mathbb{R}^d$ with covariances $r^{(i,l)}(j)=E[X_1^{(i)}X_{1+j}^{(l)}]$, and $f:\mathbb{R}^d\to\mathbb{R}$ with $Ef(X_1)^2<\infty$ and Hermite rank $q\ge 1$ with respect to

X (orthogonal to all polynomials of degree $< q$, not to some of degree q). If $\sum_{j\in\mathbb{Z}} |r^{(i,l)}(j)|^q < \infty$ for all i, l, then $n^{-1/2}\sum_{k\le n}(f(X_k) - Ef(X_k)) \Rightarrow N(0,\sigma^2)$ with $\sigma^2 = \mathrm{Var} f(X_1) + 2\sum_{k\ge 1}\mathrm{Cov}(f(X_1), f(X_{1+k})) \in [0,\infty)$.

Proof of Proposition 8. Take $d = 2$, $X_k = (X_a(k), X_b(k))$ and, for $c \in \mathbb{R}^3$, $f_c(x) = c_1(x_a^2 - 1) + c_2(x_b^2 - 1) + c_3(x_a x_b - \rho_{ab})$. f_c is a centred polynomial of degree 2 without linear part, so it is orthogonal to every polynomial of degree ≤ 1 (odd Gaussian moments vanish); if it were orthogonal to every polynomial of degree 2 it would be orthogonal to itself, so either $f_c(X_1) = 0$ a.s. (then the partial sums vanish) or its Hermite rank is 2. The condition $\sum_h |\gamma_{ab}(h)|^2 < \infty$ holds because the covariances are absolutely summable and bounded. By the theorem and the Cramér–Wold device, $\sqrt{n}(\tilde\gamma_{aa} - 1, \tilde\gamma_{bb} - 1, \tilde\gamma_{ab} - \rho_{ab}) \Rightarrow N(0, V)$, where, by Isserlis' theorem, $V(ab, cd) = \sum_h [\gamma_{ac}(h)\gamma_{bd}(h) + \gamma_{ad}(h)\gamma_{bc}(h)]$ (as in Lemma 16). Centring by the sample means changes each entry by $\bar X_a \bar X_b$, and $\sqrt{n}\bar X_a \bar X_b \to 0$ in probability since $E(\sqrt{n}\bar X_a)^2 \le \kappa_1$ and $\bar X_b \to 0$; Slutsky's lemma applies. The delta method with gradient $(-\rho/2, -\rho/2, 1)$ gives $\sqrt{n}(\hat r - \rho) \Rightarrow N(0, W)$, $W = V(ab,ab) - \rho[V(ab,aa) + V(ab,bb)] + \frac{\rho^2}{4}[V(aa,aa) + 2V(aa,bb) + V(bb,bb)]$, which equals $\sum_h \gamma_{aa}(h)\gamma_{bb}(h)$ for an independent pair (this is B_{ab} when all products $\gamma_{aa}(h)\gamma_{bb}(h)$ are nonnegative, and at most B_{ab} in general). Then $\sqrt{n}\,\operatorname{atanh}\hat r \Rightarrow N(0, W_{ab})$ under the null and $P(|\operatorname{atanh}\hat r| > z_c/\sqrt{n-3}) \to 2\bar\Phi(z_c/\sqrt{W_{ab}})$, which exceeds the nominal α/m in the limit whenever $W_{ab} > 1$. □

Proof of Proposition 9. $\mathrm{Cov}(X_u, X_v) = \gamma^Y_{uv}(0) + b_u b_v$ and $\gamma^Y_{uv} = 0$ for non-relatives (Proposition 13); divide by the standard deviations (`prop21_corr_std`). For relatives the sum is nonzero except on a Lebesgue-null set. All counts equal d, so every edge scores $1/2$ (`sOf_saturated`). The rate: apply Theorem 19 with target "all pairs" and $x = y = cn$, c small enough that $\delta \le \kappa_{\min}/3$ and $\tau_n \le \kappa_{\min}/3$ eventually; $|\hat s - 1/2| \le 1/2$ always (`abs_sOf_sub_half_le`). The onset: by the asymptotic normality of $\operatorname{atanh}\hat r$, the declaration probability crosses $1/2$ where $\tanh(z_c/\sqrt{n-3}) = |\kappa|$, i.e. at $n_{1/2}$ (`ncrit_eq_iff`). □

Proposition 20 (conditioning)**.** *Let $Z = \sqrt{r}\,C + \sqrt{1-r}\,e$ with reliability $r = \mathrm{corr}(Z,C)^2$, π_{uv} the partial correlation given Z, and ρ^Y the driver-free correlation. Then*

$$\pi_{uv} = \frac{\rho^Y_{uv}\sqrt{(1-\ell_u)(1-\ell_v)} + (1-r)\,\mathrm{sgn}(b_u b_v)\sqrt{\ell_u \ell_v}}{\sqrt{(1 - r\ell_u)(1 - r\ell_v)}}.$$

At $r = 1$ the conditional relatives are the summary relatives and Theorem 5 is restored. For non-relatives with $\ell_u, \ell_v \in (0,1)$ and $b_u b_v \neq 0$, $|\pi_{uv}|$ is strictly decreasing in r and vanishes only at $r = 1$; fewer than half of the non-relative pairs are declared only if $r > r_{\mathrm{crit}} = 1 - \frac{\tau'(1-\ell)}{\ell(1-\tau')}$, $\tau' = \tanh(z_c/\sqrt{n'-4})$ (equal shares). If the driver reaches nodes through different filters, a scalar proxy leaves a nonzero residual even at $r = 1$. The reliability is identified by $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$, the identity behind the three-indicator identification rule for one-factor models (Bollen, 1989, Ch. 7).

Lean: `prop31_std`, `cor32`, `absPi_strictAntiOn`, `absPi_eq_zero_iff`, `rcrit_iff`, `thm35_bracket`, `pc_lipschitz`, `prop36_ar1_ne_zero`, `prop36_rank_one`, `prop38_ratio`.

Proof of Proposition 20. $\mathrm{Cov}(X_u, Z) = b_u\sqrt{r}$ and $\mathrm{Var} Z = 1$; the partial covariance is $\gamma^Y_{uv} + (1-r)b_u b_v$ and the partial variance $\mathrm{Var} Y_u + (1-r)b_u^2$; divide and standardise (`prop31`, `prop31_std`; positive $\mathrm{Var} Y_u$ is required). At $r = 1$, $\pi = \rho^Y$ (`cor32`). For non-relatives $|\pi_{uv}| = \sqrt{\ell_u\ell_v}\sqrt{f_u f_v}$ with $f_w(r) = (1-r)/(1-r\ell_w)$ and $f_w'(r) = (\ell_w - 1)/(1-r\ell_w)^2 < 0$ (`absPi_strictAntiOn`, `absPi_eq_zero_iff`; both need $b_u b_v \neq 0$). Bounds $(1-r)\sqrt{\ell_u\ell_v} \le |\pi| \le (1-r)\sqrt{\ell_u\ell_v}/(1-\ell_{\max})$ (`absPi_lower`, `absPi_upper`). r_{crit} solves $(1-r)\ell/(1-r\ell) = \tau'$ (`rcrit_iff`, `rcrit_example`). The finite-sample bracket uses the Lipschitz bound $|g(a',b',c') - g(a,b,c)| \le L(q)\max(|a'-a|, |b'-b|, |c'-c|)$ for $g(a,b,c) = (a - bc)/\sqrt{(1-b^2)(1-c^2)}$, $|b|, |c| \le q < 1$, $L(q) = 1/(1-q^2) + 4q/(1-q^2)^2$ (`pc_lipschitz`, `thm35_bracket`), combined with Lemma 17 for (X, Z). Multi-filter driver: for $D_v(t) = b_v C(t - k_v)$ with C an AR(1) with coefficient ϕ and $Z = C(t)$,

$$\pi_{uv} = \frac{b_u b_v(\phi^{|k_u - k_v|} - \phi^{k_u + k_v})}{\sqrt{(s_u^2 - b_u^2\phi^{2k_u})(s_v^2 - b_v^2\phi^{2k_v})}},$$

nonzero for $k_u, k_v > 0$, $\phi \in (0,1)$, $b_u b_v \neq 0$ (`prop36_ar1`, `prop36_ar1_ne_zero`); in general $\pi_{uv} \propto \mathrm{Cov}(D_u - P_Z D_u, D_v - P_Z D_v)$ vanishes for all non-relative pairs iff the residual driver components are uncorrelated, which holds for a rank-one driver with $Z = F$ (`prop36_pcov`, `prop36_zero_iff`, `prop36_rank_one`). A proxy that contains summary relatives of u, v adds the terms $-c_Z(b_u q_v + b_v q_u) - q_u q_v$ to the numerator (`prop37_pcov`, `prop37_bound`). Triad identity: $\rho_{uZ}\rho_{vZ}/\rho_{uv} = r$ and, for $r > 0$, $\ell_u = \rho^2_{uZ}/r$ (`prop38_ratio`, `prop38_loading`). □

E. Simulation Checks of F1–F3

All with fixed seeds; full tables are in the supplementary material. **F1** (5 processes: in-forests with $d = 5$ and $d = 10$, self-loop persistence $\lambda \in \{0.5, 0.9, 0.97\}$, and a near-white-noise case; $n = 200$–$20{,}000$; 1,000 replications per cell; nominal, oracle-effective and fixed thresholds). The bound was never violated in 105 cells. It is loose: with a fixed threshold $\rho_{\min}/2$, it reaches 0.9 at about 3–13 times the n at which the simulated probability does, and much later with Bonferroni thresholds, where the Chernoff step loses the Mills-ratio factor. The empirical $n\,\mathrm{Var}(\hat{r})/W$ lies in $[0.93, 1.11]$. With the nominal test, $P(\hat{R} = R)$ plateaus at 0.94, 0.22, 0.03 and 0.27, as Proposition 8 predicts (predicted vs simulated false-positive rates, e.g. 0.33 vs 0.32–0.33 at $\lambda = 0.9$). **F2** ($\ell \in \{0.02, 0.1, 0.3, 0.64\}$): the asymptotic detection curve matches the simulated fraction of declared non-relatives within 0.02 for $n \geq 200$; predicted $n_{1/2} = 1{,}079$ for $\ell = 0.1$, simulated 0.479 at $n = 1{,}000$. **F3**: predicted $n_{\mathrm{crit}} = 472$ ($\ell = 0.64$, $r = 0.9$) gives a simulated declared fraction 0.495 at $n = 471$; over 38 cells the asymptotic prediction is within 0.03. With travel-time lags $(0, 1, 2, 4, 8)$, $\phi = 0.9$, $r = 1$, the declared fraction still rises to 0.63 at $n = 10{,}000$, as the multi-filter part of Proposition 20 predicts.

F. Lean Formalisation: Map and Findings

Lean 4.35.0 and Mathlib v4.35.0; 3,963 lines in 16 files. The build log (clean rebuild plus the incremental rebuild that added `Observed.lean`), the `#print axioms` output for each of the 148 declarations in the axiom report (all `[propext, Classical.choice, Quot.sound]`) and a grep showing no `sorry`/`admit`/`axiom` are in the supplementary material.

What is and is not certified. Certified: the exact and stationary results of Section 3, the loop-free exact edge value, the accuracy reading, the algebra of the estimand $\tilde{s}$ (Appendix H), and the deterministic and algebraic parts of Sections 4 and 5. Not certified: the ER limits (only the deterministic witness lemmas), the loop-free limit, periodicity and stationary value, the lag-weighted extension, and every probability bound, limit theorem and genericity statement.

Breakdown of the 148 declarations (counted from the Lean files): 29 headline theorems, i.e. Lean counterparts of numbered paper statements (25 for Section 3, the loop-free edge value and the accuracy reading; 4 for Proposition 22); 43 supporting graph lemmas (ancestry, roots, distance, source SCCs, 5 ER witness lemmas, 6 estimand lemmas); 65 algebraic identities and inequalities of the aggregation and finite-sample files (correlation perturbation, Chernoff algebra, trace and Toeplitz identities, partial correlation formulas, threshold locations; several are elementary, e.g. $|\hat{s} - \frac{1}{2}| \leq \frac{1}{2}$); 11 examples (10 concrete graphs, 1 numerical r_{crit} instance).

statement	Lean names	status
Lemmas 10 and 11	unrolled_reach_iff, unrolled_reach_iff_dist, roots_unrolled_iff, desc_root_unrolled_iff, isRel_unrolled_iff	checked
Theorem 1	edge_score, edge_score_dist, strictCond_iff_dist	checked
Proposition 4	rho_unrolled_eq, mT_le_iff, rho_unrolled_large_t	checked
Theorem 2	sRho_unrolled_exact, sRho_unrolled_bounds, tendsto_sRho_unrolled, tendsto_sRho_GT	checked
Corollary 3	tendsto_half_iff, crossEdges_eq_empty_tfae, sRho_unrolled_ge_of_cross, reisach_theorem4, limitValue_dag	checked
Theorem 5	rhoBurn_eq, sBurn_eq, rhoZ_eq, sZ_eq, sOf_window, relSet_eq_iff_src, mem_upEdges_iff	checked
Corollary 6	sBurn_eq_half_iff, sBurn_eq_half_iff_common_anc, sZ_eq_half_iff, upEdges_eq_empty_tfae, stat_half_of_strong, stat_half_of_outForest, inForest_isRel_iff, inForest_up_iff, statValue_inForest, statValue_inForest_gt_iff	checked
Table 1	collider_limit, fork_limit, chain3_limit, collider_stat, fork_stat, chain3_stat	checked
self-loop remark	edgeLoops_limit, edgeNoLoops_sRho	checked
Theorem 24 (ancestors, relatives, edge value)	isRel_unrolled_iff_NL, reduce_ancestor, relSet_unrolled_eq_iff_NL, edge_score_NL	checked
Corollary 25	sRho_unrolled_eq_NL, sRho_unrolled_ge_half_NL	checked
Proposition 28 (witness $u = j$, fresh parent)	strictNL_of_closed_walk, strictNL_of_fresh_parent, sRho_unrolled_eq_one_NL	checked
loop-free examples (Table 10)	colliderNL_sRho, edgeLoopHead_sRho, edgeNoLoops_sRho	checked
eventual periodicity, limit, same-SCC edges, Theorem 30	–	paper / enumerated

statement	Lean names	status
Proposition 7	witness lemmas `upEdge_witness`, `mem_upEdges_of_witness`, `mem_upEdges_of_disjoint_anc`, `upEdges_subset_confluenceNL`, `card_upEdges_le_confluenceNL`	paper proof (Appendix N.1); lemmas checked
Proposition 31	–	paper only
Proposition 22 (a)–(d)	`relObs_subset_of_edge`, `score_obsEdge`, `sTilde_eq`, `sTilde_ge_half`, `upObs_subset_upEdges`, `sTilde_gt_half_iff`, `exists_upEdge_of_sTilde_gt_half`, `sTilde_eq_half_of_no_up`, `mem_upObs_of_witness`, `upObs_univ`	checked
Proposition 41 (oracle sort accuracy; optimality)	`accRule_sortRule`, `accRule_sortRule_rho`, `accRule_le_sOf`, `accRule_le_sRho`, `edgeAcc_sortRule_strict`, `rho_not_gt_of_edge`	checked
Proposition 13	–	paper only
Theorem 14(a)	`fejer_identity` (counting identity)	partial
Theorem 14(b)	`longrun_factor`, `longrun_inv`, `selfloop_sum_lt_one`	partial
Theorem 14(c,d)	–	paper only
Lemma 15	`phi_le`, `chernoff_logmgf_le`, `chernoff_choice`, `sqrt_two_v_x`	algebra only
Lemma 16	`quad_M`, `trace_MC`, `lemma12_trace`, `lemma12_fejer`, `avg_toeplitz_le_tsum`, `lemma12_le_B`, `lemma12_diag`, `lemma12_diag_le_D`	algebra only
Lemma 17	`centred_cov_eq`, `centred_dev_offdiag`, `centred_dev_diag`, `lemma13_eps`, `lemma13_count`	deterministic part
Lemma 18	`corr_perturb`	checked
Theorem 19	`decide_eq_target`, `decide_eq_support`, `thm15_det`, `explicit_y`, `sOf_cntDecl_congr`	deterministic part
Proposition 8	–	asymptotic (cited CLT)
Proposition 9	`prop21_corr`, `prop21_corr_std`, `sOf_saturated`, `sOf_const`, `abs_sOf_sub_half_le`, `thm22_const`, `ncrit_iff`, `ncrit_eq_iff`	deterministic part
Proposition 20	`pcorr_eq_pcov`, `pcorr_eq_corr_resid`, `prop31`, `prop31_std`, `cor32`, `prop33`, `prop33_abs`, `absPi_lower`, `absPi_upper`, `absPi_le_kappa`, `absPi_lt_kappa`, `absPi_eq_zero_iff`, `absPi_strictAntiOn`, `absPi_equal`, `rcrit_iff`, `rcrit_eq_iff`, `rcrit_example`, `pc_lipschitz`, `thm35_bracket`, `prop36_pcov`, `prop36_zero_iff`, `prop36_rank_one`, `prop36_ar1_pcov`, `prop36_ar1_pvar`, `prop36_ar1`, `prop36_ar1_ne_zero`, `prop37_pcov`, `prop37_bound`, `prop38_ratio`, `prop38_loading`	population part checked

Modelling choices. Summary graphs are relations on a finite type with a self-loop hypothesis; anc and desc are reflexive-transitive closures ("a path exists iff a walk exists" is the one convention adopted without proof); G_T with $T = n + 1$ slices lives on $V \times \mathrm{Fin}(n + 1)$; source SCCs are represented by their member nodes (equality of SCC sets as equality of node unions is elementary and not formalised). In the finite-sample files, zero-mean variables are vectors of an arbitrary real inner-product space and lag-0 covariance is the inner product; independence enters only as vanishing inner products.

Findings. No statement was false. Formalisation added four missing nondegeneracy hypotheses to our earlier finite-sample statements: (i) the strict claims of Proposition 20 need $b_u b_v \neq 0$; (ii) $\ell_u = \rho_{uZ}^2 / r$ needs $r > 0$; (iii) nonvanishing of the multi-filter residual needs $b_u b_v \neq 0$; (iv) the partial correlation needs positive partial variances ($\mathrm{Var} Y_u > 0$ suffices). It also showed that one hypothesis of Proposition 4 and one of Lemma 18 are redundant, and that the in-forest statement needs acyclicity, not only out-degree at most one.

G. Real-Data Details

CausalRivers (Stein et al., 2025). Study 1 (P1–P3): the benchmark's connected-subgraph sampler, seeds 7000–7029, East Germany and Bavaria, $d \in \{5, 10\}$, 120 subgraphs, 791 induced edges; block means at 12 h and 24 h; nominal Fisher-z. Study 2 (P4–P6): seeds 8000–8029, 120 subgraphs, 1,376 non-relative pairs; proxy Z is the mean of standardised stations of

the region outside the subgraph and outside 2 undirected river hops (429–615 stations); primary test on partial correlations with a residual Bartlett n_{eff} ($K = 100$ steps at 12 h, 50 at 24 h); median n_{eff} 139/113 (raw) and 231/211 (residual). Before conditioning, 0.98 (nominal) and 0.87 (effective) of non-relative pairs tested dependent. Median proxy contamination by river relatives was 2.6% (maximum 72%), a case of the contaminated-proxy term in Proposition 20. Post hoc and without verdict: conditioning on $k = 1, 3, 5, 10$ principal components of the leave-out stations gave dependent fractions $0.745, 0.605, 0.576, 0.491$ (12 h), with accuracy 0.580 and Spearman 0.16 at $k = 10$.

Causal Chambers (Gamella et al., 2025). Candidate selection (recorded before analysis) compared Causal Chambers, CausalRivers with measured precipitation (rejected: the driver reaches each gauge through its own catchment filter, so by Proposition 20 a scalar proxy cannot remove it), a semi-synthetic cross-sectional manufacturing benchmark (rejected: not real, not time series) and fMRI with physiological regressors (rejected: no ground truth). Wind tunnel: runs `actuators_random_walk_2-16` (run 1 excluded before analysis: truncated); 16 random variables, 26 edges, 63 non-relative pairs per run; block means over 10 rows, first differences, standardisation, $n = 999$; effective-n Fisher-z with Bartlett $K = 20$. Light tunnel: `actuators_white`, 20 variables, 39 edges, ten segments of 2,000 steps, levels. Predicted values come from the package's published ground-truth graph with constant parameters removed. Secondary analyses (no verdicts): nominal n gives slightly more false positives (0.048), as Proposition 8 predicts; levels instead of increments gave 0.054, contradicting our own secondary prediction that levels of random walks would produce many spurious dependences; without block means ($D = 1$) relative recovery drops to 0.18 and Spearman to 0.10. In the light tunnel, polariser angles act on two sensors through Malus' law, a nonlinearity with near-zero linear correlation; as stated in the plan, these relative pairs were never detected, without affecting orientation.

H. The Real-Data Estimand $\tilde{s}$

The theorems of Section 3 concern s_ρ of an unrolled DAG. The real-data studies estimate something else: lag-0 dependences between the observed series of one long record. This section defines the population quantity they target and proves what it does and does not say.

Definition 21 (estimand)**.** Let $G_{\text{S}} = (V, E_{\text{S}})$ be a summary graph (no loop assumption) and $O \subseteq V$ the observed nodes. For $v \in O$ let $\tilde{\rho}_O(v) = |\text{rel}_S(v) \cap O|$ (relatives computed in the full graph G_{S}, so unobserved nodes act as latent common ancestors). Let $E_O = \{(i,j) \in E_{\text{S}} : i, j \in O,\ i \neq j\}$ be the induced non-self-loop edges and, for $E_O \neq \emptyset$, $\tilde{s}(G_{\text{S}}, O) = \frac{1}{|E_O|} \sum_{(i,j)\in E_O} s_\rho\big(\tilde{\rho}_O(i), \tilde{\rho}_O(j)\big)$, with the edge score of Section 2. Edges of the latent projection that are not edges of G_{S} (paths through unobserved nodes) are not scored.

Why this is the target. Under the SVAR model of Appendix C, Proposition 13 gives, for almost every parameter, $\Gamma(0)_{uv} \neq 0$ iff $\text{anc}_S(u) \cap \text{anc}_S(v) \neq \emptyset$ iff $v \in \text{rel}_S(u)$, for every observation scheme. So a perfect lag-0 dependence oracle on the observed series declares exactly $\text{rel}_S(v) \cap O$, and the relatives sort of the observed counts, averaged over E_O, is $\tilde{s}$. (Lag-0 faithfulness is the generic case of Proposition 13; it can fail for specific parameters, e.g. through Malus-law nonlinearities.) Relation to Theorem 5: with $O = V$, every window node of the stationary unrolled DAG has $\rho_W(v(t)) = T|\text{rel}_S(v)|$, so $\tilde{s}(G_{\text{S}}, V)$ is the stationary value computed over non-self-loop edges only.

Proposition 22 (sign relation)**.** *Let $E_O \neq \emptyset$ and $E_O^{\uparrow} = \{(i,j) \in E_O : \text{rel}_S(i) \cap O \neq \text{rel}_S(j) \cap O\}$. (a) Every edge of E_O scores $1/2$ or 1, and $\tilde{s} = \frac{1}{2} + |E_O^{\uparrow}|/(2|E_O|) \geq \frac{1}{2}$. (b) $E_O^{\uparrow} \subseteq E_{\uparrow}$. Hence $\tilde{s} > \frac{1}{2}$ iff $E_O^{\uparrow} \neq \emptyset$, and then some induced edge is an up-edge of G_{S} ($\text{Src}(i) \neq \text{Src}(j)$); if no induced edge is an up-edge, $\tilde{s} = \frac{1}{2}$. (c) $(i,j) \in E_O$ belongs to $E_O^{\uparrow}$ iff some observed w lies in $\text{rel}_S(j) \setminus \text{rel}_S(i)$, i.e. shares an ancestor with j but none with i; in particular an observed other parent k of j with $\text{anc}_S(k) \cap \text{anc}_S(i) = \emptyset$ suffices. (d) With $O = V$, $E_V^{\uparrow} = E_{\uparrow} \setminus \{\text{self-loops}\}$, so $\tilde{s}(G_{\text{S}}, V) - \frac{1}{2}$ and $s_\rho^W - \frac{1}{2}$ have the same sign. (e) The converse of (b) fails: for the collider $a \to c \leftarrow b$ with loops and $O = \{a, c\}$, the only induced edge (a, c) is an up-edge, but $\text{rel}_S(a) \cap O = \text{rel}_S(c) \cap O = \{a, c\}$, so $\tilde{s} = 1/2$.*

Lean: (a)–(d): `relObs_subset_of_edge`, `score_obsEdge`, `sTilde_eq`, `sTilde_ge_half`, `upObs_subset_upEdges`, `sTilde_gt_half_iff`, `exists_upEdge_of_sTilde_gt_half`, `sTilde_eq_half_of_no_up`, `mem_upObs_of_witness`, `upObs_univ` (file `Observed.lean`); (e) by hand.

Proof. (a) $i \in \text{anc}_S(j)$ gives $\text{rel}_S(i) \subseteq \text{rel}_S(j)$ (Reisach et al., 2026, Thm. 1), hence the same after intersecting with O; nested sets have equal or strictly ordered sizes. (b) If the observed parts differ, the full sets differ, and by Theorem 5 that is $\text{Src}(i) \neq \text{Src}(j)$. (c) By (a) the observed sets are nested, so they differ iff the larger one has an observed element outside the smaller; $w \in \text{rel}_S(j) \setminus \text{rel}_S(i)$ means $\text{anc}_S(w) \cap \text{anc}_S(j) \neq \emptyset = \text{anc}_S(w) \cap \text{anc}_S(i)$; k itself is such a w. (d) is (b) with

$O = V$ and $\mathrm{Src}(i) \neq \mathrm{Src}(j)$ iff $\mathrm{rel}_S(i) \neq \mathrm{rel}_S(j)$ (Theorem 5); $s_\rho^W > \frac{1}{2}$ iff $E_\uparrow \neq \emptyset$, and self-loops are never in $E_\uparrow$. (e) $\mathrm{rel}_S(a) = \{a, c\}$, $\mathrm{rel}_S(c) = \{a, b, c\}$; b is unobserved. □

What is and is not implied. For river subgraphs (in-forests) an induced edge (i, j) counts iff an observed station lies upstream of j on a branch other than i's (in an in-forest $\mathrm{rel}_S(v) = \mathrm{anc}_S(v) \cup \mathrm{desc}_S(v)$ and $\mathrm{desc}_S(j) \subseteq \mathrm{desc}_S(i)$), so a confluence in the induced subgraph whose other branch is observed is sufficient. $\tilde{s}$ is a statement about population lag-0 relatives of a stationary process; it is not s_ρ of the finite unrolled DAG, of the induced subgraph's unrolled DAG, or of the latent projection, and it says nothing about lagged dependence. A deviation of the estimate from $\tilde{s}$ is a failure of estimation (F1–F3) or of lag-0 faithfulness, not evidence against Theorems 1 and 2.

I. Time-Stamped Analysis Plans: Process, Prior Exposure, Timeline, Deviations

Process. The four analysis plans, all analyses and the results were produced by AI coding agents (Section 7, AI-use statement) within about six hours of one day. Each plan was committed to version control before its study's data were loaded for any dependence statistic in this project. Commit timestamps are set by the committing client; no third-party registry (e.g. OSF) was used, and the hosting service's push records, the stronger evidence, can be supplied to the area chair on request. The gap between plan and results was short: about 3 min 41 s for plan 3 and 3 min 36 s for plan 4 (each including a run of about two minutes). We therefore call these *internally time-stamped analysis plans*; earlier drafts called them pre-registrations. Commit hashes are withheld for anonymity and will be provided with the de-anonymised version.

Prior exposure to CausalRivers (disclosure). One day before plan 1, a separate exploratory project in the same repository (a test of contemporaneous-edge discovery on aggregated CausalRivers; commits dated 2026-11-03) had analysed CausalRivers with the same regions (East Germany, Bavaria), at resolutions 6 h, 12 h and 24 h (the studies here use 12 h and 24 h), and with the same data loaders and subgraph sampler that the plan-1 code reuses (with new seeds). It ran correlation, VAR, PCMCI+, VAR-LiNGAM and DYNOTEARS baselines and recorded a hidden-driver diagnostic: at 24 h the VAR(1) residual correlation among sibling stations was 0.51, the same as among unrelated non-adjacent stations in the same subgraph (0.51), and below adjacent pairs (0.65); it concluded that shared regional forcing is large and contemporaneous dependence is not specific to the river edges. This was known to the agents (and recorded in the repository) before plan 1 was written. Plan 1 nevertheless predicted that the relatives sort would orient river edges above chance; it failed. Because the strong shared dependence had been observed beforehand, the F2 saturation account of that failure (and the F3 account of study 2) is *explanatory*, not a prediction made blind to the data. The Causal Chambers data and the E1/E2 held-out sets had not been analysed before their plans.

Times below are commit timestamps relative to the first plan commit (T_0).

time	event
$T_0 + 0{:}00$	plan 1 committed: targets T1–T3, predictions P1–P3, kill rule
$T_0 + 0{:}14$	theory of Sections 3 and 4, enumeration checks, operationalisation of P1–P3 (before data)
$T_0 + 0{:}17$	study 1 results (P2, P3 fail; kill rule triggered)
$T_0 + 1{:}05$	plan 2 committed: F1–F3 targets, Lean target, predictions P4–P6, kill rule
$T_0 + 1{:}18$–$1{:}31$	Lean formalisation of Section 3 and algebraic parts of Section 4
$T_0 + 1{:}38$	finite-sample theory, simulations, operationalisation of P4–P6 (before data)
$T_0 + 1{:}44$	study 2 results (P4, P6 fail; kill rule triggered)
$T_0 + 4{:}58$	plan 3 committed: candidate scoping, graph-only predictions, P7–P9 (before any dependence statistic)
$T_0 + 5{:}02$	study 3 results (P7, P8 pass; P9 fails; replication passes)
$T_0 + 5{:}14$–$5{:}24$	Lean formalisation of the finite-sample lemmas; errata
$T_0 + 5{:}45$	manuscript draft committed (with placeholders for E1/E2)
$T_0 + 5{:}50$	plan 4 committed: E1 (test of the theorem against eight naive predictors), E2 (lagged relatives), theory addendum A1–A3 and B1–B3, graph-only predictions, design check; development runs on the study-3 data only; no dependence statistic of any held-out dataset computed
$T_0 + 5{:}54$	E1/E2 results committed as planned (E1 passes 8/8 on the primary graph; Q7, Q8, Q10 pass, Q9 fails); out-of-sample F checks; run logs

Deviations (all recorded at the time). (1) Plan 1 said "theorem-predicted s_ρ" without naming the version; the stationary version was fixed in writing before data were loaded (the finite version would predict $> 1/2$ for every subgraph, including chains). (2) Self-loops are excluded from the predicted average because the estimator cannot orient them; this does not

change which side of $1/2$ the prediction is on. (3) Only 6 of 120 study-1 subgraphs lacked a confluence; the plan set no minimum size for that group, and its trivial pass is not interpreted. (4) Plan 2 did not say which test defines "dependent" in P4; the effective-n test was fixed as primary before data, and the nominal test fails P4 as well. (5) In study 3, a planned triad-reliability check was not applicable (the measured driver reaches only four mutually related barometers) and is not interpreted. (6) The P9 failure is close to the threshold and its CI contains 0.3; as fixed in the plan, the point estimate decides. (7) Plan 4 (E1/E2): no threshold, unit, seed, K, L, D, graph or node set changed after the results were seen. A stray cache file committed with plan 4 was removed in the results commit (no content change). The F-check tolerance of ± 0.03 is the one fixed in plan 4; the citris "predicted accuracy" is undefined (no full-set edge of G_{lin} is predicted 1). The constant-$1/2$ comparison and the weak-component-size comparison are post hoc. The choice of G_{lin} as the primary graph was fixed in plan 4, together with the secondary analysis on the published graph G_D, which is the one on which the theorem fails against the sensor indicator on strictly ordered edges.

J. Experiment E1: Discriminating Test Against Naive Predictors

Status and provenance. Specified in plan 4 (Appendix I); the plan was committed before any dependence statistic of any held-out dataset was computed. Done before it: file hashes, column and row counts, regeneration of the `citris_1` input protocol from its published generator (it matches the recorded inputs exactly), graph-only predictions, and development runs on the study-3 data only. All numbers below come from the committed result files of the experiment folder (`results4.json`, `summary4.txt`); none is estimated or rounded beyond the printed digits.

Why `citris_1`. Every published Causal Chambers graph is bipartite except the pressure-control configuration, whose only experiment varies no independent input, so we rejected it. The light-tunnel `citris_1` series (100,000 steps about 1.8 s apart) is generated by a time-series structural model among the actuators: red is a reflected random walk, green moves halfway towards red and blue a quarter of the way towards green (each plus a uniform step), pol_1 is a random walk and pol_2 moves towards or away from pol_1 according to $\text{sign}(\text{blue} - \text{red})$. With the sensors this is a chain of length 3, red $\to$ green $\to$ blue $\to$ sensors. The secondary data are snapshot (i.i.d.) structural models among the actuators (`buchholz_1_obs`; `scm_{1,2,4,5}_reference`, including a chain and a collider) and the wind-tunnel regime-jump runs (bipartite; disagreements arise only from latent-node ties).

Linear-zero lemma (the refinement). Write $\mathcal{C}$ for the colour block (red, green, blue, ir_1, vis_1, ir_2, vis_2, ir_3, vis_3, current) and $\mathcal{Q}$ for the polariser block (pol_1, pol_2, angle_1, angle_2). *B1 (proved for the actuators).* Conditionally on the flag and the colour paths, the law of the polariser paths is invariant under negation, because every update (uniform draw on a symmetric interval, move towards or away with a sign fixed by the colours, reflection at ± 90, rounding to 0.1) commutes with negation. *B2 (assumed).* The first-two-layer sensors depend only on the colours and independent noise; ir_3 and vis_3 depend on the polarisers only through an even function of $\theta_1 - \theta_2$ (Malus' law with aligned zero marks); each angle sensor is affine in its polariser plus independent noise. This is a physical assumption that the design documents do not let us check. *B3.* Under B1 and B2 every variable of $\mathcal{C}$ has zero cross-covariance, at every pair of times, with every variable of $\mathcal{Q}$ (a product of an even and an odd function of the polariser path has conditional mean zero). The six real edges red $\to \text{pol}_2$, blue $\to \text{pol}_2$ and $\text{pol}_k \to \text{ir}_3$, vis_3 therefore carry no linear dependence. The relatives that a correlation-based estimator targets are those of the *linear-dependence graph* G_{lin} (the design graph G_D without these six edges), which has two weak components, each with a single source (red; pol_1). By Corollary 6, every edge inside a block has predicted value $1/2$ for every node set; the six cross-block edges are compared by block size: red $\to \text{pol}_2$ and blue $\to \text{pol}_2$ are predicted 0 (anti-sorted) and $\text{pol}_k \to \text{ir}_3$, vis_3 are predicted 1 for the full set. Faithfulness inside the blocks is assumed, as in Proposition 13. A zero-mark offset or a non-affine angle sensor would break B2 and move the prediction towards G_D.

Metric and tests. For each edge instance (the full node set plus 20 subgraphs per unit) the observed value is $o \in \{0, \frac{1}{2}, 1\}$ from the estimated counts, and the loss is $|o - p|$ for the predicted value p. For each competitor take the instances where it disagrees with the theorem; $\Delta = \text{loss(theory)} - \text{loss(competitor)}$, pooled, with a unit-level exact sign-flip test (1,024 flips for 10 units) and Holm correction over eight competitors, and a unit-bootstrap CI. E1a (prerequisites): non-relatives declared < 0.5 and relatives detected ≥ 0.8. E1b: $\Delta \leq -0.10$ and Holm $p \leq 0.05$ for all eight. E1c: the same restricted to instances whose own pair is declared dependent ($\Delta < 0$). Separation was to be claimed iff E1a, E1b and E1c all pass on `citris_1`. Kill rules: K1 (E1a fails: no claim), K2 (a competitor is significantly better: no claim against it), K3 (E1b passes and E1c fails: attributed to power-induced ties). None was triggered on the primary analysis.

Observed values given the theorem's prediction: predicted $1/2$ (1,365 instances) observed exactly $1/2$ in 99.2%; predicted 1 (187 instances, polariser $\to \text{ir}_3$, vis_3) observed 1 in 99.5%; predicted 0 (110 instances, red and blue $\to \text{pol}_2$) observed 0 in

Table 5. E1 primary, `citris_1`, graph G_{lin}, lag 0, 1,662 instances. Δ [95% CI]; all rows: 10/10 units, Holm $p = 0.0078$. "Exact": share of instances whose predicted value equals the observed one. "Spearman": agreement of the mean predicted edge value with observed accuracy over 200 subgraphs.

predictor	Δ [CI]	Δ, detected edges	exact	Spearman
theorem	—	—	0.992	0.962
sensor indicator	-0.502 [-0.507, -0.496]	-0.500	0.203	0.408
ancestors, full graph	-0.515 [-0.519, -0.510]	-0.500	0.147	0.334
ancestors, subgraph	-0.509 [-0.513, -0.505]	-0.500	0.120	0.121
in-degree, full graph	-0.515 [-0.520, -0.511]	-0.500	0.180	0.349
in-degree, subgraph	-0.510 [-0.514, -0.505]	-0.500	0.143	0.105
depth, full graph	-0.515 [-0.519, -0.510]	-0.500	0.147	0.334
depth, subgraph	-0.510 [-0.514, -0.505]	-0.500	0.114	0.113
every edge 1	-0.533 [-0.536, -0.530]	-0.500	0.115	undefined

99.1%. The detected-edge subset has 1,338 instances, all observed exactly $1/2$. Prerequisites: non-relatives declared 0.000 [0, 0] (40 pairs per unit), relatives detected 0.969 [0.945, 0.992] (51 pairs per unit).

Strict subset and the power safeguard. On the 297 instances where the theorem predicts 0 or 1 it beats all eight ($\Delta = -0.539$ for the sensor indicator, -0.948 for ancestors and depth on the full graph, -0.991 for "every edge 1"; Holm $p = 0.0078$, 10/10 units). There, an estimator without power would favour the competitors (the "all counts tied" baseline is $+0.443$ to 0), so this part cannot come from power-induced ties.

Published graph G_D (secondary). Applying the theorem to the unmodified design graph, relatives detection is 0.659 [0.643, 0.675] (75 relative pairs per unit) and non-relatives declared 0.000. Overall the theorem still beats all eight ($\Delta = -0.323$ for the sensor indicator; -0.424 for ancestors, depth and "every edge 1"; -0.420 and -0.417 for in-degree; 10/10 units, Holm $p = 0.0078$), and its exact-match rate is 0.714. But on the 442 edges where G_D predicts strict order (1), the theorem matches 0.222 and the sensor indicator 0.319; the sensor indicator is significantly better ($\Delta = +0.453$ [0.410, 0.492], theorem better in 0/10 units, reverse Holm $p = 0.002$). A constant-$1/2$ rule (post hoc) matches 0.815 of all instances, against 0.714 for the theorem on G_D and 0.992 on G_{lin}; mean $|o - 1/2| = 0.092$ against 0.166 for the theorem on G_D. The Spearman statistic is 0.341 (theorem on G_D) against 0.408 (sensor indicator). This is why the claim is restricted to the linearly dependent relatives.

Lagged estimator on `citris_1` (secondary). With $L = 5$ and G_{lin}: detection 0.976 [0.953, 0.992], non-relatives declared 0.000, 8/8 passed ($\Delta = -0.505$ to -0.535), theorem exact match 0.998, Spearman 0.996.

Table 6. E1 on i.i.d. structural-model sets (no verdict; secondary). Competitors' exact-match range over the eight predictors. Units: `scm_1` 5, the others 10 each (45 pooled).

set	instances	theorem exact	competitors exact	Δ range	result
`buchholz_1_obs`	1,712	0.791	0.328–0.375	-0.499 to -0.496	pass 8/8
`scm_1_reference`	867	1.000	0.314–0.392	-0.500	not met[†]
`scm_2_reference`	1,650	0.812	0.324–0.408	-0.443 to -0.426	pass 8/8
`scm_4_reference` (chain)	2,267	1.000	0.000–0.088	-0.500	pass 8/8
`scm_5_reference` (collider)	2,238	1.000	0.464–0.495	-0.500	pass 8/8
pooled (45 units)	8,734	0.923	0.283–0.322	-0.489 to -0.487	pass 8/8, Holm $p = 8 \times 10^{-5}$

[†]Five units: the smallest attainable Holm-adjusted p is $8/32 = 0.25$.

On the snapshot sets, relatives detected: 0.933 (`buchholz_1_obs`), 1.000 (`scm_1`), 0.965 (`scm_2`), 1.000 (`scm_4`, `scm_5`); non-relatives declared 0.000 wherever non-relative pairs exist (`scm_4` has none). The chain (`scm_4`) gives $1/2$ on all 2,267 instances (observed exactly $1/2$ every time); the collider gives 1 on edges into the collider and its sensors and $1/2$ elsewhere (all 2,238 instances exact). Pooled, the theorem's strict predictions number 2,768 (observed value 1 on 2,403, $1/2$ on 349, 0 on 16: exact 0.868); on them only the sensor indicator disagrees often (390 instances, $\Delta = -0.5$, 35/35 units) and the other competitors coincide with the theorem. The pooled Spearman is 0.913 for the theorem and -0.138 for the sensor indicator. Post hoc, a constant-$1/2$ rule matches 0.688 of the pooled instances against 0.923.

Wind-tunnel regimes (tertiary, 64 units, no verdict). The same test passes against all eight in each configuration. $D = 10$, $L = 0$: non-relatives declared 0.014, relatives detected 0.592, $\Delta = -0.203$ [-0.227, -0.179] in 63/64 units, exact match

0.539 against 0.432; detected-edge subset $\Delta = -0.169$. $D = 1$, $L = 30$: 0.003, 0.398, $\Delta = -0.225$ $[-0.250, -0.197]$ in 62/64 units, exact 0.443 against 0.353; detected-edge subset $\Delta = -0.134$. $D = 1$, $L = 0$: $\Delta = -0.218$ (64/64 units), exact 0.411 against 0.326; $D = 10$, $L = 3$: $\Delta = -0.218$, exact 0.533 against 0.411. Detection of 0.33–0.59 means that part of the advantage is power-induced, as plan 4 stated in advance. Post hoc, the constant-$1/2$ rule has mean absolute error 0.311 against 0.363 for the theorem at $D = 1$, $L = 30$ (and 0.313 against 0.294 at $D = 10$, $L = 0$), and matches 0.378 (0.375) of the instances against 0.443 (0.539).

A trivial rule that E1 did not test (post hoc). Each weak component of G_{lin} has a single source (red; pol_1). By the proof of Corollary 6, every node of such a component is reached from that source, so its relatives are the whole component, and for every node set the predicted counts are the numbers of observed nodes in the endpoints' weak components. The theorem's prediction therefore coincides on all 1,662 instances with the rule "$1/2$ iff both endpoints lie in the same weak component of G_{lin}, otherwise compare the observed sizes of the two components". That rule was not among the eight competitors, so E1 cannot separate the theory from it; the only data in this appendix where the source structure matters beyond component membership are the snapshot sets with several sources per component, notably the collider `scm_5`.

What E1 does and does not show. It shows that, on one multi-layer light-tunnel series, the estimated relatives counts are tied along the whole chain red $\to$ green $\to$ blue $\to$ sensors and between the polariser block and the angle sensors, and strictly ordered (including against the causal direction) on red, blue $\to \text{pol}_2$ — and that the source-component rule of Theorem 5 and Corollary 6 predicts this while eight naive rules do not. It does not show that the theorem predicts relatives under the published graph; the faithfulness of the graph to linear dependence fails here by design. Most of the separation is in the ties, which naive rules cannot express, and on G_{lin} the prediction equals the weak-component-size rule. The evidence is one device in a favourable regime, and the wind tunnel, where detection is limited by F1, gives weaker support.

K. Experiment E2: Lagged Relatives

Theory (plan 4, addendum A). For $L \in \{0, 1, \dots\}$ let the lag-L relatives of u be u and every v with $\gamma_{uv}(h) \neq 0$ for some $|h| \leq L$. *A1 (proved).* Under the independent-innovations assumption behind Proposition 13, $\text{rel}_0 \subseteq \text{rel}_L \subseteq \text{rel}_\infty \subseteq \text{rel}_S$; for a VAR model all are equal for almost every parameter value; and Theorem 5, Corollary 6 hold verbatim with "relatives" read as rel_L under lag-L faithfulness. Lag-0 faithfulness implies lag-L faithfulness; the converse fails (a white-noise source s with an edge $s \to j$ and no self-loop at s has $\gamma_{sj}(0) = 0 \neq \gamma_{js}(1)$). *Estimator.* This is a Bonferroni max-over-lags variant of the classical cross-correlation tests of independence between two series (Haugh, 1976 cross-correlates prewhitened residuals and uses a portmanteau statistic; we use effective-n Fisher-z statistics without prewhitening). Declare u, v relatives iff $\max_{|h| \leq L} |\operatorname{atanh} \hat{r}_{uv}(h)| \sqrt{\hat{n}_{\text{eff},uv}(h) - 3} > z_c(L)$, with $\hat{n}_{\text{eff},uv}(h) = (n - |h|)/\hat{B}_{uv}$ and $z_c(L) = \Phi^{-1}(1 - \alpha/(2m(2L+1)))$, a Bonferroni correction over the m pairs and the $2L + 1$ lags; $L = 0$ is exactly the study-3 test. *A2 (proved for finite n).* For independent stationary Gaussian a, b the null variance of the lag-h cross-covariance has the same bound at every h, so one Bartlett factor serves all lags. *A3 (deterministic part proved, constants sketched).* The recovery bound of F1 holds with $\rho_{\min}$ replaced by $\rho^*_{\min,L} = \min_{\text{relative pairs}} \max_{|h| \leq L} |\rho_{ab}(h)| \geq \rho_{\min}$ (the gain) and a factor $2L + 1$ in the failure probability and the Bonferroni count (the cost): for 105 pairs at $\alpha = 0.01$, z_c rises from 3.90 ($L = 0$) to 4.35 ($L = 3$) and 4.80 ($L = 30$). A lagged correlation must therefore exceed the lag-0 correlation by about 1.12 ($L = 3$) or 1.23 ($L = 30$) times to help.

Design. Held-out data: the wind-tunnel regime-jump runs (`regime_jumps_single`, `regime_jumps_multi`), $2 \times 32 = 64$ regimes of 10,000 steps, in the same archive as the study-3 data but never analysed in study 3. Load_in, load_out, pot_1, pot_2 follow independent mirrored random walks within a regime; the hatch and sensor parameters are constant within a regime. The first 100 rows of each regime are dropped. Graph: 20 edges without the hatch (19 predicted 1, $\text{pot}_1 \to \text{signal}_1$ predicted $1/2$); 54 non-relative and 51 relative pairs; 1,280 subgraphs (10 of size 5 and 10 of size 8 per unit). Primary: $D = 1$ (first differences), $L = 30$ (about 9 s) against $L = 0$; secondary: block means with $D = 10$ and $L = 3$ against $L = 0$. Success required Q7–Q10 all to pass at the primary configuration. Kill rules: Q7 fails (saturation) or Q10 fails (lags do not help detection, and the lagged estimator is not recommended). Plan 4 states that E2 is not a re-analysis of P9 and that P9's verdict is final.

Detection of the 51 relative pairs: $D = 1$: 0.328 [0.312, 0.345] at lag 0 and 0.398 [0.383, 0.412] with $L = 30$; $D = 10$: 0.592 [0.584, 0.599] and 0.585 [0.576, 0.592] with $L = 3$ (lags reduce detection: the Bonferroni cost exceeds the gain). On `citris_1` with $L = 5$ against $L = 0$: 0.976 against 0.969, difference $+0.008$ [0, 0.024], so Q10 fails there; detection is already nearly saturated at lag 0. **Verdict.** Q9 fails at the primary configuration, so E2 does not succeed; no kill rule is triggered (Q7 and Q10 pass). Lags raise detection at fine resolution without adding false positives, as A1 and A3 predict.

Table 7. E2, 64 held-out regimes, 95% unit-bootstrap CIs. Q8: accuracy on the 1,216 instances predicted 1. Q10: relatives detected, lagged minus lag 0 (non-relatives declared difference in the last column).

configuration	Q7	Q8	Q9	Q10	non-rel. diff.
primary $D{=}1$, $L{=}30$	0.003 [0.001, 0.005] pass	0.553 [0.527, 0.579] pass	0.243 [0.193, 0.294] **fail**	$+0.069$ [0.060, 0.079] pass	$+0.001$ [-0.001, 0.002]
secondary $D{=}10$, $L{=}3$	0.011 [0.008, 0.014]	0.711 [0.692, 0.729]	0.202 [0.152, 0.253] fail	-0.007 [-0.012, -0.003] fail	-0.002 [-0.004, -0.0003]
lag 0, $D{=}10$ (replication of study 3)	0.014 [0.010, 0.018]	0.742 [0.722, 0.760]	0.202 [0.145, 0.259] fail	—	—
lag 0, $D{=}1$	0.002 [0.001, 0.004]	0.552 [0.532, 0.570]	0.120 [0.066, 0.176] fail	—	—

The study-3 estimator (lag 0) rerun on these held-out regimes fails the graded criterion again (0.202 at $D = 10$, 0.120 at $D = 1$): the P9-type failure repeats on held-out data. P9's original verdict (0.267, failed) stands; E2 is not presented as rescuing it. The development check of plan 4 showed that, on the study-3 data with the same estimator, drawing different subgraph seeds moves the P9 statistic from 0.267 to 0.338. We report this as variability of the statistic (about 0.07 from the subgraph draw alone); it is a development run, not a verdict, and it does not change the P9 outcome.

Table 8. Detection rate per edge over 64 regimes: share of regimes in which the pair was declared dependent.

edge	$D{=}1$, $L{=}0$	$D{=}1$, $L{=}30$	$D{=}10$, $L{=}0$
load_in → rpm_in	0.859	0.875	1.000
load_in → rpm_out	0.000	0.172	0.094
load_in → current_in	1.000	1.000	1.000
load_in → current_out	0.000	0.000	0.047
load_out → rpm_in	0.000	0.016	0.000
load_out → rpm_out	0.844	0.969	1.000
load_out → current_out	1.000	1.000	1.000
load_out → current_in	0.000	0.000	0.000
load_in → pressure_intake	0.000	0.094	0.953
load_out → pressure_intake	0.000	0.031	0.906
load_in → pressure_upwind	0.562	0.984	1.000
load_out → pressure_upwind	0.016	0.812	1.000
load_in → pressure_downwind	0.734	0.953	1.000
load_out → pressure_downwind	0.750	1.000	1.000
$\text{pot}_1 \to \text{signal}_1$	0.359	0.156	0.969
$\text{pot}_1 \to \text{signal}_2$	0.266	0.109	0.969
$\text{pot}_2 \to \text{signal}_2$	0.000	0.000	0.266
pot_1, load_in, load_out → mic	0.000	0.000	0.016 (pot_1), 0.000

Lags recover the pressure edges at fine resolution (load_in → upwind pressure $0.562 \to 0.984$, load_out → upwind pressure $0.016 \to 0.812$) but lose the pot edges to the larger threshold ($0.359 \to 0.156$ and $0.266 \to 0.109$); every microphone edge, $\text{pot}_2 \to \text{signal}_2$ and the cross-current edges stay undetected. Even at its best ($D = 10$, lag 0) wind-tunnel detection (0.592) is far below that of the light tunnel, so lags are not the main cause of the F1 shortfall.

Table 9. Out-of-sample F-theory checks: the first half of each unit predicts the second half. Predicted detection is a bracket $[\bar{p}_{\max}, \bar{p}_{\text{ind}}]$ when lags are used. Pass if the observed value lies within 0.03 of the bracket (criterion fixed in plan 4).

configuration	predicted detection	observed	non-rel. declared, pred. / obs.	accuracy, pred. / obs.
wind $D{=}1$, $L{=}0$	0.250	0.242	0.007 / 0.003	0.573 / 0.590
wind $D{=}1$, $L{=}30$	[0.344, 0.426]	0.332	[0.021, 0.074] / 0.003	0.504 / 0.506
wind $D{=}10$, $L{=}0$	0.548	0.533	0.014 / 0.008	0.649 / 0.632
wind $D{=}10$, $L{=}3$	[0.536, 0.554]	0.524	[0.016, 0.027] / 0.008	0.628 / 0.618
`citris_1`, $L{=}0$	0.935	0.945	0.007 / 0.000	undefined
`citris_1`, $L{=}5$	[0.940, 0.988]	0.945	[0.010, 0.045] / 0.000	undefined

All six checks pass. With lags the observed detection sits at or slightly below $\bar{p}_{\max}$, consistent with strongly correlated lag statistics; the independence formula overstates both detection and false positives, as the theory said it would. For `citris_1` the predicted accuracy is undefined because no full-set edge of G_{lin} is predicted 1 inside a block.

L. Summary Graphs without Self-Loops

Here $G_S = (V, E_S)$ is an arbitrary directed graph with $|E_S| \geq 1$; no loop assumption is made. B_n is the boolean n-th power of the adjacency matrix ($B_0 = I$), $\Lambda(u, v) = \{n : B_n[u, v] = 1\}$ the set of lengths of walks from u to v, and u is *fresh* if it has no incoming edge in E_S (a self-loop is an incoming edge). Status: Theorem 24 and Corollary 25 are machine-checked; the rest of this section is proved on paper and checked by enumeration, as marked.

Lemma 23 (ancestors and roots without loops)**.** *$u(a) \in \mathrm{anc}(v(t))$ iff $a \leq t$ and $t - a \in \Lambda(u, v)$. A node $u(a)$ is a root of G_T iff $a = 1$ or u is fresh.*

Lean: `unrolled_reach_iff` *(already free of the loop hypothesis).*

Proof. Every edge of G_T raises time by one, so paths from $u(a)$ to $v(t)$ are the lifts of walks of length exactly $t - a$, and every walk lifts. Without loops one cannot pad. A node $u(a)$ with $a \geq 2$ has a parent iff some edge (w, u) exists. □

Theorem 24 (edge value without loop assumption)**.** *Let $(i, j) \in E_S$ and $e_t = (i(t), j(t{+}1))$, $1 \leq t < T$. Then $s_\rho(e_t) \in \{\frac{1}{2}, 1\}$ independently of T, and $s_\rho(e_t) = 1$ iff $\mathrm{Strict}_{ij}(t)$ holds, where $\mathrm{Strict}_{ij}(t)$ is*

$$\exists u :\ t \in \Lambda(u, j),\ t - 1 \notin \Lambda(u, i) \quad \text{or} \quad \exists \text{ fresh } u,\ \exists n \in \{1, \ldots, t\} :\ n \in \Lambda(u, j),\ n - 1 \notin \Lambda(u, i).$$

Lean: `isRel_unrolled_iff_NL`, `reduce_ancestor`, `relSet_unrolled_eq_iff_NL`, `edge_score_NL`.

Proof. As in Theorem 1, $\mathrm{rel}(i(t)) \subseteq \mathrm{rel}(j(t{+}1))$, and the score is 1 iff $\mathrm{roots}(j(t{+}1)) \setminus \mathrm{roots}(i(t)) \neq \emptyset$ (Reisach et al., 2026, Thm. 1, Lemma 1). By Lemma 23 an element of the difference is $u(a)$ with ($a = 1$ or u fresh), $n = t{+}1{-}a \in \Lambda(u, j)$ and $u(a) \notin \mathrm{roots}(i(t))$. The case $a = t + 1$ would make j fresh, impossible since j has the parent i; so $a \leq t$ and, by the lemma again, $u(a)$ is not a root of $i(t)$'s ancestry iff $n - 1 = t - a \notin \Lambda(u, i)$. The branch $a = 1$ is $n = t$ with no condition on u; the other is a fresh u with $1 \leq n \leq t$. (The Lean proof avoids Lemma 1: the extra relative is $u(0)$ in the first case and a node of a fresh u in the second, and `reduce_ancestor` pushes any common ancestor back along parents to slice 0 or to a fresh node.) □

Corollary 25. *For every summary graph and $T \geq 2$, $s_\rho(G_T) = \frac{1}{2} + \frac{1}{2(T-1)|E_S|} \sum_{(i,j) \in E_S} N_T(i, j)$ with $N_T(i, j) = \#\{t < T : \mathrm{Strict}_{ij}(t)\}$; in particular $s_\rho(G_T) \geq 1/2$. With a loop at every node no node is fresh, $\Lambda(u, j) = [d(u, j), \infty)$, and $\mathrm{Strict}_{ij}(t)$ becomes $\exists u : d(u, j) \leq t \leq d(u, i)$, i.e. Theorem 1.*

Lean: `sRho_unrolled_eq_NL`, `sRho_unrolled_ge_half_NL`.

Exhaustive check of Theorem 24 against brute-force relatives: $p = 2, 3$ ($T = 2, \ldots, 9$), all 65,535 graphs with $p = 4$ ($T = 3, 5, 8$), and random graphs with $p = 5, 6$; 0 mismatches in 120, 3,577, 196,605, 8,928 and 4,494 (graph, T) cases (enumerated, not part of the Lean proof).

Proposition 26 (limit exists and is rational; paper proof, enumerated)**.** *(B_n) is eventually periodic: there are $n_0, \pi \geq 1$ with $B_{n+\pi} = B_n$ for $n \geq n_0$. Hence $L = \lim_T s_\rho(G_T) = \frac{1}{2} + \frac{1}{2|E_S|} \sum_e \delta_e$ with $\delta_e = 1$ if some fresh u has $n \geq 1$, $n \in \Lambda(u, j)$, $n - 1 \notin \Lambda(u, i)$, and otherwise δ_e the density, over one period of $t > n_0$, of t with some u, $B_t[u, j] = 1$, $B_{t-1}[u, i] = 0$; and $|s_\rho(G_T) - L| \leq (n_0 + \pi + 1)/(2(T - 1))$.*

Proof. B_{n+1} is a function of B_n and there are finitely many boolean matrices. A fresh witness is monotone in t and has its first n at most $n_0 + 1 + \pi$ (the set $\Lambda(u, j) \setminus (\Lambda(u, i) + 1)$ is eventually periodic), so such an edge is strict for all $t \geq n_0 + 1 + \pi$. Without a witness, $\mathrm{Strict}_{ij}(t)$ depends for $t > n_0$ only on $t \bmod \pi$. Counting strict $t < T$ gives the error bound; average over edges. □

Checked on 1,115 graphs (all $15 + 511$ graphs with $p = 2, 3$, 390 random with $p = 4$, 199 random with $p = 5, 6$): the exact value at $T = 600$ is within the bound and $(L - \frac{1}{2})\, 2|E_S|\, \pi$ is an integer in every case.

Proposition 27 (edges inside an SCC; paper proof, enumerated)**.** *Let $(i, j) \in E_S$ with j reaching i and no fresh witness. Then $\mathrm{Strict}_{ij}(t)$ fails for every $t \geq n_0 + 1$.*

Proof. Fix u; put $f(n) = B_n[u, j]$, $h(n) = B_{n-1}[u, i]$. The edge gives $h(n) \leq f(n)$. If m is the length of a walk $j \to i$ and $c = m + 1$, then $f(n) \leq h(n + c)$. Take $k \geq 1$ with kc a multiple of π. For $n \geq n_0 + 1$, $h(n) \leq f(n) \leq h(n + c) \leq \cdots \leq h(n + kc) = h(n)$, so $f(n) = h(n)$ for every u. □

Proposition 28 (crossing edge into a cyclic node)**.** *If (i, j) is crossing and j lies on a cycle with period g (gcd of the closed-walk lengths at j), then* $\mathrm{Strict}_{ij}(t)$ *holds for every $t \in \Lambda(j, j)$, which contains all large multiples of g, so $\delta_e \geq 1/g$, and $\delta_e = 1$ if $g = 1$. A fresh parent $u \neq i$ of j makes every layer strict.*

Lean: `strictNL_of_closed_walk`, `strictNL_of_fresh_parent`, `sRho_unrolled_eq_one_NL`.

Proof. Take $u = j$: $t \in \Lambda(j, j)$ and $t - 1 \notin \Lambda(j, i) = \emptyset$. Closed-walk lengths at j form a numerical semigroup in $g\mathbb{Z}$, which contains all large multiples of g. □

Corollary 29 (loop-free versions of the loop-dependent statements)**.** *(a) If every weakly connected component of G_{S} is strongly connected (any loop pattern, any periods), then $L = 1/2$ and $s_\rho(G_T) - \frac{1}{2} \leq n_0/(2(T-1))$ (paper proof from Proposition 27; Reisach et al.'s Theorem 4 without loops). (b) The converse and the closed form are false without loops (Table 10). (c) If every crossing edge has a head on a cycle, then $L > 1/2$ iff $E_\times \neq \emptyset$.*

Table 10. Loop-free examples (L: finite limit, S: stationary). Lean: `edgeNoLoops_sRho`, `colliderNL_sRho`, `edgeLoopHead_sRho`; the others are computed values (example scripts in the supplement).

summary graph	without loops	with loops
single edge $0 \to 1$	$s_\rho(G_T) = 1/2$ for all T	$L = 2/3$, $S = 1/2$
chain $0 \to 1 \to 2$	$L = 1/2$	$L = 0.7$
2-cycle $\{0, 1\}$ plus $0 \to 2$	$L = 1/2$	loop at 2 only: $L = 5/8$
collider	$L = S = 1$	$L = S = 0.7$
edge, loop at head only	$L = S = 1$	–

For an acyclic loop-free graph, $L(u, v)$ is finite, $B_n = 0$ for $n \geq p$, and $L = \frac{1}{2} + \#\{e : \text{fresh witness}\}/(2|E_{\mathrm{S}}|)$ (paper proof); for a loop-free in-forest an edge (i, j) has a fresh witness iff j has another parent, so $L = \frac{1}{2} + (m - n_1)/(2m)$ with $m = |E_{\mathrm{S}}|$ and n_1 the number of nodes with one parent, which equals the stationary value below. On 362 random acyclic loop-free graphs $L = S$ in all cases, and the in-forest formula holds on 298 in-forests.

Theorem 30 (stationary value without loops; (a), (c) proved, (b) sketched)**.** *Let G_{S} be arbitrary, $G_{\mathbb{Z}}$ the infinite-past unrolled DAG on $V \times \mathbb{Z}$, and $\Delta(v, w) = \{m - n : \exists u,\ n \in \Lambda(u, v),\ m \in \Lambda(u, w)\}$. Then $\mathrm{rel}_{G_{\mathbb{Z}}}(v(t)) = \{w(t + d) : d \in \Delta(v, w)\}$ (proved), the edge $(i(t), j(t+1))$ is strict iff $\Delta(j, w) + 1 \neq \Delta(i, w)$ for some w (proved; always $\Delta(i, w) \subseteq \Delta(j, w) + 1$), and $S^{\mathbb{Z}} = \frac{1}{2} + |E^{\mathbb{Z}}_{\uparrow}|/(2|E_{\mathrm{S}}|)$. (a) Outside $|d| < n_0 + \pi$, $\Delta(v, w)$ is π-periodic, so $E^{\mathbb{Z}}_{\uparrow}$ is detected with witnesses of $|d| \leq n_0 + 2\pi$ (proved). (b) In a finite window after a burn-in $B \geq 2n_0 + 3\pi + 1$ the windowed value equals $S^{\mathbb{Z}} + O(K_0/T)$, $K_0 = n_0 + 2\pi$ (*sketched*). (c) $S^{\mathbb{Z}} = 1/2$ whenever every weak component is strongly connected, whatever its periods (a node colouring* lab *with $\Delta(v, w) = \mathrm{lab}(w) - \mathrm{lab}(v) + g\mathbb{Z}$) (proved). With loops everywhere Δ is $\mathbb{Z}$ or empty and this is Theorem 5.*

Brute-force relatives in G_{B+T} ($B = 40$, $T = 90$, edges at distance ≥ 30 from the window ends) agree with $E^{\mathbb{Z}}_{\uparrow}$ on every edge of all $15 + 511$ graphs with $p = 2, 3$ and 248 random graphs with $p = 4$ (enumerated). Not formalised in Lean: periodicity, Proposition 27, Corollary 29(a), Theorem 30.

M. Lagged Unrollings

Theorems 1, 2 and 5 and their Lean proofs are for VAR(1)-shaped unrollings: every summary edge acts at lag one. Here every non-loop summary edge $e = (i, j)$ carries a lag $\tau_e \in \{1, \ldots, \tau_{\max}\}$ (a pair may carry several lags, each a separate edge), self-loops act at lag one, and G_T has the edges $(i(t), j(t+\tau_e))$, $1 \leq t \leq T - \tau_e$. Let $d_\tau(u, v)$ be the smallest total lag of a path from u to v (0 if $u = v$, ∞ if none); $d_\tau(u, v) \leq (p-1)\tau_{\max}$ when finite. Status: paper proof, not machine-checked and not enumerated. Contemporaneous (lag-0) edges are excluded: they break the time-monotonicity used below.

Proposition 31 (lag-weighted version)**.** *(a) $u(s) \in \mathrm{anc}(v(t))$ iff $d_\tau(u, v) \leq t - s$. (b) The copy $(i(t), j(t+\tau_e))$ scores 1 iff some u has $d_\tau(u, j) - \tau_e + 1 \leq t \leq d_\tau(u, i)$, and $1/2$ otherwise. (c) Crossing edges score 1 in every copy; an edge inside an SCC has at most $(p-1)\tau_{\max}$ strict copies, so $\lim_{T\to\infty} s_\rho(G_T) = \frac{1}{2} + |E_\times|/(2|E_{\mathrm{S}}|)$ with $|s_\rho(G_T) - L| = O(p\tau_{\max}/T)$. (d) With a latent burn-in $B \geq (p-1)\tau_{\max}$ (or an infinite past), Theorem 5 holds verbatim.*

Proof. (a) Every edge raises time by its lag, so a path from $u(s)$ to $v(t)$ projects to a walk of total lag $t - s$; conversely a path of total lag $d_\tau(u, v)$ can be padded by lag-one self-loops at u. (b) As in Lemma 11, the roots of $v(t)$ are $\{u(1) : d_\tau(u, v) \leq$

$t-1\}$ (every $u(s)$, $s \geq 2$, has the parent $u(s-1)$), and $i(t)$ is a parent of $j(t+\tau_e)$, so the score is 1 iff $j(t+\tau_e)$ has a root $u(1)$ with $d_\tau(u,j) \leq t+\tau_e-1$ that $i(t)$ lacks, $d_\tau(u,i) > t-1$. (c) For a crossing edge take $u=j$: $d_\tau(j,j)=0 \leq t+\tau_e-1$ and $d_\tau(j,i)=\infty$. Inside an SCC, $d_\tau(u,i)<\infty$ iff $d_\tau(u,j)<\infty$, so a strict copy needs $t \leq d_\tau(u,i) \leq (p-1)\tau_{\max}$. Each edge has $T-\tau_e$ copies; average. (d) By (a), $w(s) \in \mathrm{rel}(v(t))$ iff some u has $d_\tau(u,v) \leq t-1$ and $d_\tau(u,w) \leq s-1$; for window nodes $t-1, s-1 \geq B \geq (p-1)\tau_{\max}$ this is $\mathrm{anc}_S(v) \cap \mathrm{anc}_S(w) \neq \emptyset$, and the proof of Theorem 5 applies. □

N. Random Summary Graphs: Derivations and Checks

All graphs here are a non-loop random graph plus a self-loop at every node, so $|E_{\mathrm{S}}| = p+m$ with m the number of non-loop edges, and for acyclic graphs $|E_\times| = m$.

In-forests (proved identities). Let the non-loop part be an in-forest with m edges and n_1 nodes with exactly one parent. By Corollaries 3 and 6, $L = \frac{1}{2}+\frac{m}{2(m+p)}$, $S = \frac{1}{2}+\frac{m-n_1}{2(m+p)}$, $L-S = \frac{n_1}{2(m+p)}$ (the last checked to machine precision on 18 random forests). For a sequence of in-forests with parent-count distribution π and mean μ: $L \to \frac{1}{2}+\frac{\mu}{2(1+\mu)}$ and $S \to \frac{1}{2}+\frac{\mu-\pi_1}{2(1+\mu)}$. Exact computation at $p = 20{,}000$ (6 seeds each): uniform recursive in-tree ($\pi_1 = 1/4$, $S \to 11/16 = 0.6875$, measured 0.68739 ± 0.00026); preferential attachment (measured $\pi_1 = 0.1656$, $S = 0.70859 \pm 0.00028$); uniform random labelled tree rooted and oriented to the root (Poisson(1) parents, $\pi_1 \to e^{-1}$, $S \to \frac{1}{2}+(1-e^{-1})/4 = 0.65803$, measured 0.65783 ± 0.00020). In all three $L = 0.74999$.

Directed ER: summary. $G(p, c/p)$, $m/p \to c$. The finite limit needs the fraction of non-loop edges inside an SCC ($\to x^2$, the two endpoints lie in the giant SCC iff the ancestor tree of i and the descendant tree of j are infinite); the stationary value needs the fraction of up-edges ($\to F_\uparrow$). Both are proved in Appendix N.1 below. Large c: $1-x \sim e^{-c}$, $L-\frac{1}{2} \sim e^{-c}$, $S-\frac{1}{2} \sim e^{-c}/2$.

Layered DAG (sketch; L proved given edge concentration). M layers of width w, each node of layer $l+1$ receiving an edge from each node of layer l with probability c/w; as $w \to \infty$ with $\mu = c(M-1)/M$, $L \to \frac{1}{2}+\frac{\mu}{2(1+\mu)}$ and $S \to \frac{1}{2}+\frac{\mu(1-e^{-c})}{2(1+\mu)}$. For fixed w and deep M the stationary value falls below this (observed 0.7824 against 0.7878 at $w = 50$, $M = 400$, $c = 2$); this correction is not derived.

Table 11. Exact computation (means ± s.d. over seeds; SCCs by scipy and source sets by bitsets) against the limits of Proposition 7, proved in Appendix N.1.

c	p	L exact	L formula	S exact	S formula
0.5	20000	0.6666 ± 0.0011	0.6667	0.5658 ± 0.0014	0.5656
1.0	20000	0.7486 ± 0.0008	0.7500	0.6557 ± 0.0013	0.6580
1.5	50000	0.6975 ± 0.0020	0.6981	0.6159 ± 0.0018	0.6163
2.0	50000	0.6214 ± 0.0022	0.6217	0.5629 ± 0.0014	0.5632
3.0	100000	0.5434 ± 0.0010	0.5433	0.5215 ± 0.0006	0.5214
5.0	20000	0.5059 ± 0.0005	0.5058	0.5028 ± 0.0003	0.5029

L at $c = 1$ converges as p grows: 0.7466 ($p = 2000$), 0.7486 ($2 \cdot 10^4$), 0.74976 ($2 \cdot 10^5$), 0.74992 (10^6); at $c = 2$, 0.62162 ± 0.00023 at $p = 10^6$ against 0.62170. The giant-SCC fraction matches x^2 (0.6353 against 0.6349 at $c = 2$). The second fixed point matters: with $d = 1$ the formula would give $S = 0.6312, 0.5766, 0.5246$ at $c = 1.5, 2, 3$, outside the error bars, whereas the fixed points $d = 0.485, 0.221, 0.061$ give 0.6163, 0.5632, 0.5214. Further exact rows: $c = 1.5$, $p = 10^5$, 32 graphs, $L = 0.69819 \pm 0.00024$ (formula 0.69810); $c = 2$, $p = 3.2 \cdot 10^5$, $L = 0.62170 \pm 0.00012$ (0.62170). The Barak–Erdős phase-transition paper cited by Reisach et al. (2026) concerns monotone-path reachability at the dense scale $\log n/n$ and is not used here.

N.1. Directed Erdős–Rényi summary graphs: full proof of Proposition 7

Status. Everything in this subsection is proved on paper. The only inputs used without proof are textbook facts (F1)–(F3) below, checked against the author's online version of van der Hofstad (2016) (theorem numbers are those of that version and may differ in print). Lemma 35 is the standard local-coupling argument for sparse random graphs and is written concisely. The deterministic witness lemmas are machine-checked; no probabilistic step is. No human probabilist has yet checked this subsection (see the AI-use statement).

Limits and their order. For a fixed summary graph, Theorem 2 gives $\lim_{T\to\infty} s_\rho(G_T) = L(G_{\mathrm{S}}) = \frac{1}{2}+|E_\times|/(2|E_{\mathrm{S}}|)$,

and Theorem 5 gives $s_\rho^W = \frac{1}{2} + |E_\uparrow|/(2|E_S|)$ for every T. For $G_S \sim D(p, c/p)$ we take the limit $T \to \infty$ at fixed p first (this defines $L(G_S)$) and then let $p \to \infty$; for the stationary value no T-limit is needed. The regime of fixed T with $p \to \infty$ is not covered (the bound of Theorem 2 is vacuous for $T \le p$).

Model and notation. $D(p, c/p)$: each ordered pair (i, j), $i \neq j$, is a non-loop edge independently with probability $q = c/p$, and every node has a self-loop (loops change neither reachability nor Src). m is the number of non-loop edges, $|E_S| = p + m$, N the number of non-loop edges inside an SCC, so $|E_\times| = m - N$. $\mathrm{out}(v)$, $\mathrm{in}(v)$ are the sets of nodes reachable from v, resp. reaching v (v included), T_v^{out}, T_v^{in} their sizes; $v \rightsquigarrow w$ means $w \in \mathrm{out}(v)$. $x = x(c)$ is the largest solution of $x = 1 - e^{-cx}$ (the survival probability of a Poisson(c) Galton–Watson tree), $\theta_K = \theta_K(c)$ the probability that its total progeny is at least K ($\theta_K \downarrow x$), and d the smallest fixed point in $[0, 1]$ of $f(d) = 1 + e^{-c} - e^{-c(1-x)d}$. A node is *in-isolated* if it has no non-loop parent. $\mathcal{G}$ denotes the *edge-rooted Poisson tree*: an edge (i, j); i has Poisson(c) parents and Poisson(c) further children, j Poisson(c) further parents and Poisson(c) children; every other node has Poisson(c) parents and Poisson(c) children besides the edge through which it was reached; all independent.

Textbook facts. (F1) Poisson coupling (van der Hofstad, 2016, Thm. 2.10): $d_{\mathrm{TV}}(\mathrm{Bin}(n, q), \mathrm{Poi}(c)) \le nq^2 + |nq - c|$. (F2) Chernoff: for a sum X of independent Bernoulli variables with mean μ and $\delta \in (0, 1)$, $P(X \le (1-\delta)\mu) \le e^{-\delta^2\mu/2}$. (From $Ee^{-tX} \le \exp(\mu(e^{-t} - 1))$ with $t = -\log(1-\delta)$, Markov's inequality, and $(1-\delta)\log(1-\delta) \ge -\delta + \delta^2/2$; the upper-tail version $P(X \ge \mu + s) \le e^{-s}$ for $s \ge e^2\mu$ follows in the same way.) (F3) Galton–Watson facts (van der Hofstad, 2016, Thms. 3.1, 3.16): the extinction probability is the smallest root of $\eta = e^{c(\eta-1)}$, so the survival probability is $x(c)$, positive iff $c > 1$.

Deterministic lemmas (machine-checked, `ErLemmas.lean`; no loop hypothesis). A node z is a *source node* if every ancestor of z is reachable from z. By Theorem 5, $(i, j) \in E_\uparrow$ iff some source node reaches j but not i.

(W1) If $(i, j) \in E_\uparrow$, there are $k \notin \{i, j\}$ with $k \to j$ and a source node z with $z \rightsquigarrow k$, $z \not\rightsquigarrow i$ (`upEdge_witness`).

(W2) If $i \to j$, $k \to j$, and a source node reaches k but not i, then $(i, j) \in E_\uparrow$ (`mem_upEdges_of_witness`).

(W3) If $i \to j$, $k \to j$ and $\mathrm{in}(k) \cap \mathrm{in}(i) = \emptyset$, then $(i, j) \in E_\uparrow$ (`mem_upEdges_of_disjoint_anc`).

(W5) A non-loop edge (i, j) lies inside an SCC iff $j \rightsquigarrow i$. The event $\{j \rightsquigarrow i\}$ and the set $\mathrm{out}(j)$ do not change when the edge (i, j) is added or removed (a simple path from j to i does not use it). (Trivial; not formalised.)

Explorations and forced edges. Forward exploration of $\mathrm{out}(v)$ keeps explored, active and unseen sets X_t, A_t, U_t, $a_t = |A_t|$, $\mathrm{disc}_t = t + a_t$, $X_0 = \emptyset$, $A_0 = \{v\}$. Step $t + 1$ takes the oldest active node w, reveals the pairs (w, u), $u \in U_t$, and makes the ξ_{t+1} heads of present pairs active; it stops at $T = T_v^{\mathrm{out}}$. Backward exploration of $\mathrm{in}(v)$ reveals the pairs (u, w) instead. Let F be a set of at most three ordered pairs and G^F the graph in which the pairs of F are present and all other pairs are independent Be(q). *Bookkeeping.* In a forward exploration a pair is revealed only when its tail is explored, and each node is explored once, so each pair is revealed at most once; the pairs revealed at step $t + 1$ are disjoint from those revealed before and from all pairs whose tail is not in X_{t+1}. Hence, conditionally on the history $\mathcal{H}_t$, the non-forced pairs revealed at step $t + 1$ are independent Be(q), and $\xi_{t+1} = \#\{u \in U_t : (w, u) \in F\} + \mathrm{Bin}(|U_t| - f_w, q)$ with $f_w \le 3$ the number of forced pairs from w into U_t. Thus, given $\mathcal{H}_t$, ξ_{t+1} stochastically dominates $\mathrm{Bin}(|U_t| - 3, q)$ and is dominated by $3 + \mathrm{Bin}(p, q)$. The same holds for backward explorations (with heads in place of tails), and, by the same argument, for a backward exploration run after a forward one, conditionally on the forward history, as long as it never discovers a node that the forward exploration discovered (only pairs with tail outside X_f and head explored backward are revealed, and none of them was revealed forward).

Fix $c > 1$ and $\varepsilon \in (0, 1/5)$ with $\beta = c(1 - 4\varepsilon) > 1$; put $n' = \lceil (1 - 4\varepsilon)p \rceil$, $\kappa = (\beta - 1)^2/(8\beta)$, $\delta = (\beta - 1)\varepsilon/(\beta + 1)$, $t_0 = C_3 \log p$ with $C_3\kappa \ge 8$. Call an exploration *admissible* if, conditionally on its history, every offspring count ξ_{t+1} taken while $\mathrm{disc}_t < \varepsilon p$ dominates $\mathrm{Bin}(n', q)$ and every offspring count is dominated by $3 + \mathrm{Bin}(p, q)$. By the bookkeeping, forward and backward explorations in G^F are admissible for large p (since $|U_t| - 3 > (1 - \varepsilon)p - 3 \ge n'$), and so is the backward exploration of Lemma 33 below.

Lemma 32 (dichotomy)**.** *There is C (depending on c, ε) such that for large p every admissible exploration with final size T satisfies (a) $P(K \le T < \varepsilon p) \le Ce^{-\kappa K}$ for all $K \ge 1$; (b) $P\big(T \ge t_0$ and [$\mathrm{disc}_t < \varepsilon p$ for all t, or $a_\tau < \delta p$ at $\tau = \min\{t : \mathrm{disc}_t \ge \varepsilon p\}$]$\big) \le Cp^{-8}$. Consequently, with probability at least $1 - Cp^{-7}$, every node v of G has, in both directions, $T_v < t_0$ or ($T_v \ge \varepsilon p$ and $a_\tau \ge \delta p$).*

Proof. Let ξ'_t be independent $\mathrm{Bin}(n', q)$ variables coupled so that $\xi_t \geq \xi'_t$ for $t \leq \tau$ (admissibility; after τ or after the exploration stops, ξ'_t uses external randomness), and $W_t = 1 + \sum_{s \leq t}(\xi'_s - 1)$. By induction $a_t \geq W_t$ for $t \leq \min(T, \tau)$. Since $tn'q \geq t\beta$ and $(\beta+1)/2 = (1 - \frac{\beta-1}{2\beta})\beta$, (F2) gives $P(W_t \leq (\beta-1)t/2) \leq P(\mathrm{Bin}(tn', q) \leq t(\beta+1)/2) \leq e^{-\kappa t}$, so $\mathrm{Bad}(t_1) = \{\exists t \geq t_1 : W_t \leq (\beta-1)t/2\}$ has probability at most $e^{-\kappa t_1}/(1-e^{-\kappa})$. (a) If $T < \varepsilon p$ then $\tau = \infty$, so $0 = a_T \geq W_T$, which is $\mathrm{Bad}(K)$ when $T \geq K$. (b) On $\mathrm{Bad}(t_0)^c$ with $T \geq t_0$: for $t_0 \leq t \leq \min(T, \tau)$, $a_t \geq W_t > 0$, so the exploration cannot stop before τ, hence $\tau < \infty$. At τ, $a_\tau + \tau \geq \varepsilon p$, and if $\tau \geq t_0$ also $a_\tau > (\beta-1)\tau/2$; minimising $\max((\beta-1)\tau/2, \varepsilon p - \tau)$ over τ gives $a_\tau \geq \delta p$; if $\tau < t_0$, $a_\tau \geq \varepsilon p - t_0 \geq \delta p$. $e^{-\kappa t_0} \leq p^{-8}$. The consequence is a union bound over p nodes and two directions. □

Lemma 33 (large out-sets reach large in-sets)**.** *With probability at least $1 - Cp^{-5}$: for all v, w, $T_v^{\mathrm{out}} \geq t_0$ and $T_w^{\mathrm{in}} \geq t_0$ imply $v \rightsquigarrow w$.*

Proof. Fix (v, w). Explore $\mathrm{out}(v)$ forward up to τ_f (history $\mathcal{H}_f$, discovered set $R_f = X_f \cup A_f$). Since every ξ is dominated by $3 + \mathrm{Bin}(p, q)$, $P(\max_t \xi_t \geq \varepsilon p) \leq e^{-\varepsilon p}$ for large p (F2, upper tail), so $|R_f| \leq 2\varepsilon p$ except on that event. Then explore $\mathrm{in}(w)$ backward, stopping at τ_b or when a node of $R_f \subseteq \mathrm{out}(v)$ is discovered; in the latter case $v \rightsquigarrow w$. Otherwise the backward exploration is the true exploration of $\mathrm{in}(w)$, its unseen set has at least $p - 2\varepsilon p - \varepsilon p - 3 \geq n'$ usable nodes while $\mathrm{disc} < \varepsilon p$, and by the bookkeeping it is admissible conditionally on $\mathcal{H}_f$. By Lemma 32(b) for both explorations, outside probability $2Cp^{-8}$: if $T_v^{\mathrm{out}}, T_w^{\mathrm{in}} \geq t_0$ and no connection occurred, then $|A_f|, |A_b| \geq \delta p$. A_f and A_b are disjoint, and every pair $(a, a') \in A_f \times A_b$ is unrevealed (its tail is not in X_f, its head not in X_b), so given both histories these pairs are independent $\mathrm{Be}(q)$ (or forced present): $P(\text{no edge } A_f \to A_b) \leq (1-q)^{\delta^2 p^2 - 3} \leq e^{-c\delta^2 p/2}$. An edge $a \to a'$ gives $v \rightsquigarrow a \to a' \rightsquigarrow w$. Union bound over p^2 pairs. □

Let Ω be the intersection of the events of Lemma 32 (consequence) and Lemma 33, $P(\Omega^c) \leq Cp^{-5}$. Put $\mathcal{S} = \{v : T_v^{\mathrm{out}} \geq t_0\}$, $\mathcal{T} = \{v : T_v^{\mathrm{in}} \geq t_0\}$.

Corollary 34 (structure)**.** *On Ω: $\mathcal{S} \times \mathcal{T}$ is contained in the reachability relation; every SCC other than the one containing $\mathcal{S} \cap \mathcal{T}$ has fewer than t_0 nodes; if $|\mathcal{S} \cap \mathcal{T}| \geq t_0$, this SCC equals $\mathcal{S} \cap \mathcal{T}$.*

Proof. The first claim is Lemma 33. A node $u \notin \mathcal{S} \cap \mathcal{T}$ has $T_u^{\mathrm{out}} < t_0$ or $T_u^{\mathrm{in}} < t_0$, and its SCC lies in $\mathrm{out}(u) \cap \mathrm{in}(u)$. If u were in the SCC of $\mathcal{S} \cap \mathcal{T}$, that SCC (of size $\geq t_0$) would lie in $\mathrm{out}(u) \cap \mathrm{in}(u)$, a contradiction. □

(This re-proves the part of the Karp–Łuczak structure theorem we need; the statement is recorded in the introduction of Dovgal et al., 2020.)

Lemma 35 (local concentration; standard coupling)**.** *Fix R and a function φ of the isomorphism class of the radius-R neighbourhood $B_R(e)$ of a non-loop edge e (all edges incident to nodes at undirected distance $< R$ from e, rooted at the oriented edge e), and let $\Phi_p = p^{-1}\sum_e \varphi(B_R(e))$, the sum over non-loop edges. (a) If φ is bounded, $E\Phi_p \to cE_{\mathcal{G}}\varphi$ and $\mathrm{Var}\,\Phi_p \to 0$; the same holds for node-rooted functionals with the node-rooted Poisson tree. (b) If $\varphi \geq 0$ and $\varphi(B_R(e)) \leq \mathrm{indeg}(j)$ for $e = (i, j)$, then $\limsup_p E\Phi_p \leq cE_{\mathcal{G}}[\min(\varphi, M)] + C/M$ for every M, with C not depending on p or M.*

Proof. (a) First moment: $E\sum_e \varphi(B_R(e)) = \sum_{(i,j)} q\, E[\varphi(B_R((i,j); G \cup \{(i,j)\}))]$. Explore the undirected neighbourhood of (i, j) breadth first, revealing for each explored node w all pairs (w, u) and (u, w). On $\{|B_R| \leq M\}$, each explored node has $\mathrm{Bin}(|U|, q)$ new children and $\mathrm{Bin}(|U|, q)$ new parents among the unseen nodes, $|U| \geq p - M$, which (F1) couples with $\mathrm{Poisson}(c)$ up to an error $\leq Mq^2p + cM/p$ per count, and the pairs between already seen nodes are present with total probability $\leq M^2q$. So the truncated ball is within $C(M, R)/p$ in total variation of the truncated $\mathcal{G}$. Tightness: each exploration step is dominated by $\mathrm{Bin}(2p, q)$ (resp. $\mathrm{Poisson}(2c)$), so $E|B_R| \leq (1 + 2c + o(1))^R$ and $P(|B_R| > M) \leq C^R/M$ uniformly in p; let $p \to \infty$, then $M \to \infty$. Second moment: $E\Phi_p^2 = p^{-2}\big[\sum_e + \sum_{e,f \text{ sharing a node}} + \sum_{e,f \text{ node-disjoint}}\big]$. The first two sums are $O(p)$ since $E\sum_v d(v)^2 = O(p)$. For node-disjoint e, f, explore both balls jointly in $G \cup \{e, f\}$: on {both sizes $\leq M$} they are disjoint and non-adjacent except with probability $\leq C(M, R)/p$, and then the two explorations are coupled with independent copies of $\mathcal{G}$. Hence $p^{-2}\sum_{\text{disjoint}} \to (cE_{\mathcal{G}}\varphi)^2$ and $\mathrm{Var}\,\Phi_p \to 0$. (b) Apply (a) to $\min(\varphi, M)$ and bound the rest by $E\sum_{(i,j)} \mathrm{indeg}(j)\mathbf{1}[\mathrm{indeg}(j) > M] = E\sum_j \mathrm{indeg}(j)^2\mathbf{1}[\mathrm{indeg}(j) > M] \leq p\,E[D^3]/M$, $D \sim \mathrm{Bin}(p-1, q)$, and $E[D^3] \leq c^3 + 3c^2 + c$ for all p (uniform integrability). □

Local functionals. All of the following are functions of $B_{2K+1}(e)$ (a breadth-first search of the first K nodes of a set stays within distance $K-1$), are invariant under isomorphism, and count all qualifying nodes, so no labelling enters. $A_e^K = \mathbf{1}[T_j^{\text{out}} \geq K,\ T_i^{\text{in}} \geq K]$, with $E_{\mathcal{G}} A^K = \theta_K^2$ (the descendant tree of j and the ancestor tree of i are independent Poisson Galton–Watson trees). $\Psi_K(e) = \text{cl}_1 \vee \text{cl}_2$, with $\text{cl}_1 = [T_i^{\text{in}} < K$ and j has a parent $k \notin \{i,j\}]$ and $\text{cl}_2 = [$some in-isolated s has $T_s^{\text{out}} < K$, $j \in \text{out}(s)$, $i \notin \text{out}(s)]$. $\varphi_K^{\beta}(e) = \#\{k \notin \{i,j\} : k \to j,\ T_k^{\text{in}} < K,\ T_i^{\text{in}} < K,\ \text{in}(k) \cap \text{in}(i) \neq \emptyset\} \leq \text{indeg}(j)$, with $E_{\mathcal{G}} \min(\varphi_K^{\beta}, M) = 0$ (in $\mathcal{G}$ the ancestor trees of i and of another parent of j are disjoint).

Proposition 36 (L, $c > 1$)**.** *For $c > 1$, $N/p \to cx^2$ in L^1 and in probability.*

Proof. By (W5), $N = \sum_e Z_e$ with $Z_e = \mathbf{1}[j \rightsquigarrow i]$; compare with $\tilde{N}_K = \sum_e A_e^K$. (i) $Z = 1, A = 0$: $j \rightsquigarrow i$ and ($T_j^{\text{out}} < K$ or $T_i^{\text{in}} < K$). By (W5) these events do not depend on the presence of e, so the expected number of such edges is $p(p-1)q\,P(i \in \text{out}(j), T_j^{\text{out}} < K)$ plus the same with $\text{in}(i)$, and by exchangeability of the $p-1$ nodes other than j, $P(i \in \text{out}(j), T_j^{\text{out}} < K) = E[(T_j^{\text{out}} - 1)\mathbf{1}(T_j^{\text{out}} < K)]/(p-1) \leq K/(p-1)$. Total $\leq 2cK$. (ii) $Z = 0, A = 1$: on Ω, $T_j^{\text{out}} \geq t_0$ and $T_i^{\text{in}} \geq t_0$ would give $j \rightsquigarrow i$, so T_j^{out} or T_i^{in} lies in $[K, t_0) \subseteq [K, \varepsilon p)$. The expected number of such edges is at most $\sum_{(i,j)} q\, P_{G^{\{e\}}}(\cdot) \leq cp \cdot 2Ce^{-\kappa K}$ by Lemma 32(a) in $G^{\{e\}}$; on Ω^c the count is at most m and $E[m\mathbf{1}_{\Omega^c}] \leq (Em^2)^{1/2} P(\Omega^c)^{1/2} = O(p \cdot p^{-5/2})$. So $E|N - \tilde{N}_K| \leq 2cK + 2cCpe^{-\kappa K} + o(1)$. By Lemma 35(a), $\tilde{N}_K/p \to c\theta_K^2$ in L^2; hence $\limsup_p E|N/p - cx^2| \leq 2cCe^{-\kappa K} + c|\theta_K^2 - x^2| \to 0$ as $K \to \infty$. □

The same argument for the node functional $\mathbf{1}[T_v^{\text{out}} \geq K, T_v^{\text{in}} \geq K]$ gives $|\mathcal{S} \cap \mathcal{T}|/p \to x^2$, so by Corollary 34 the giant SCC has $(x^2 + o_P(1))p$ nodes and all other SCCs fewer than $C_3 \log p$, w.h.p.

Proposition 37 (the tree computation)**.** *In $\mathcal{G}$, with Ψ_∞ defined by $T_i^{\text{in}} < \infty$ and $T_s^{\text{out}} < \infty$, $E_{\mathcal{G}} \Psi_K \uparrow P_{\mathcal{G}}(\Psi_\infty) = F_\uparrow$ as $K \to \infty$.*

Proof. Both clauses increase in K to their $K = \infty$ versions (monotone convergence). Let $A(i)$ be the event that the ancestor tree of i is infinite and $D(j)$ that the descendant tree of j is infinite (probability x each, independent). The other parents of j are Poisson(c) many. (1) Not $A(i)$: Ψ_∞ holds iff j has another parent. Clause 1 needs exactly that; clause 2 needs it too, since a path from s to j ends with an edge $k \to j$ and $k = i$ would put i into $\text{out}(s)$. Probability $(1-x)(1-e^{-c})$. (2) $A(i)$: clause 1 fails. Clause 2 needs (a) not $D(j)$, since $\text{out}(s)$ contains the descendant tree of j (probability $1-x$), and (b) another parent k of j and a chain $k = u_0 \leftarrow u_1 \leftarrow \cdots \leftarrow u_r = s$ ending at an in-isolated s, in which every u_t has only finite descendant trees below its children other than u_{t-1} ($u_{-1} = j$); such an s does not reach i, because paths into i stay in the ancestor tree of i. The probability that all other children of a node have finite descendant trees is $E[(1-x)^{\text{Poi}(c)}] = e^{-cx} = 1-x$. Let $Q(v)$ be the event that v is in-isolated or some parent u of v has all other children finite and $Q(u)$ holds, and $d^* = P(Q)$. Parents are Poisson(c) many and each is *good* (other children finite and Q) independently with probability $(1-x)d^*$, so by Poisson thinning $d^* = e^{-c} + 1 - e^{-c(1-x)d^*} = f(d^*)$. Q is the increasing union of its depth-n truncations Q_n, with $P(Q_0) = e^{-c} = f(0)$ and $P(Q_{n+1}) = f(P(Q_n))$; as f is increasing and continuous, $P(Q_n) = f^{n+1}(0)$ increases to the smallest fixed point d of f (if d' is a fixed point, $f^n(0) \leq f^n(d') = d'$). So $d^* = d$. The number of good other parents of j is Poisson($c(1-x)d$), so (b) has probability $1 - e^{-c(1-x)d}$, independently of $A(i)$ and $D(j)$. Adding (1) and (2) gives $F_\uparrow$. □

(An independent Monte Carlo of the event itself, not of the closed form, gives 0.38756 ± 0.00077 against 0.38773 at $c = 1.5$ and 0.19049 ± 0.00062 against 0.18962 at $c = 2$.)

Proposition 38 (reduction of $E_\uparrow$ to Ψ_K, $c > 1$)**.** $E\sum_e |\mathbf{1}[e \in E_\uparrow] - \Psi_K(e)| \leq (\delta_1(K) + o(1))p$ *with* $\delta_1(K) = C(c^2 + K^2)e^{-\kappa K} + o_K(1)$, *where $o_K(1) \to 0$ as $K \to \infty$ and C does not depend on K or p.*

Proof. Let $e = (i,j)$ be a non-loop edge. *Clause 2 never errs*: an in-isolated s is a source node, so $j \in \text{out}(s)$, $i \notin \text{out}(s)$ give $e \in E_\uparrow$. *Case I:* $T_i^{\text{in}} < K$. (I-a) If $e \in E_\uparrow$, (W1) gives another parent of j, so clause 1 holds: no error. (I-b) Clause 1 holds and $e \notin E_\uparrow$. Then $\mathbf{1}[\text{I-b}] \leq \sum_{k \notin \{i,j\}} \mathbf{1}[e, (k,j)$ present, $T_i^{\text{in}} < K,\ e \notin E_\uparrow]$ (a union over *all* other parents; no parent is singled out). Fix the triple (i,j,k) and force $F = \{(i,j),(k,j)\}$. By (W3), $e \notin E_\uparrow$ forces $\text{in}(k) \cap \text{in}(i) \neq \emptyset$, and by (W2) every source node reaching k reaches i. Split on T_k^{in}: (α_1) $K \leq T_k^{\text{in}} < \varepsilon p$: probability $\leq Ce^{-\kappa K}$ (Lemma 32(a), backward, in G^F). (α_2) $T_k^{\text{in}} \geq \varepsilon p$: the backward exploration of $\text{in}(k)$ explores at least $\varepsilon p/2$ nodes. When a node $y \neq j$ is explored, the pairs (u,y) with u unseen are revealed (all absent with conditional probability $\geq (1-q)^p$), and the pairs (u,y) with u already seen are never revealed by this exploration and are independent of it and of each other (distinct heads); so, given the history, y is in-isolated with probability $\geq (1-q)^p \geq e^{-c}/2$, and the number of in-isolated nodes among the first

$\varepsilon p/2$ explored nodes dominates $\mathrm{Bin}(\varepsilon p/2-1, e^{-c}/2)$. By (F2) at least K of them are in-isolated except with probability $e^{-\Omega(p)}$. Each in-isolated node of $\mathrm{in}(k)$ is a source node reaching k, hence reaches i, so lies in $\mathrm{in}(i)$; but $|\mathrm{in}(i)|<K$. So (α_2) has probability $e^{-\Omega(p)}$. (β) $T_k^{\mathrm{in}}<K$ and $\mathrm{in}(k)\cap\mathrm{in}(i)\neq\emptyset$. Summed over k this is $\varphi_K^{\beta}(e)$. The number of triples is p^3 and each has both pairs present with probability q^2, so cases (α_1), (α_2) contribute at most $c^2p(Ce^{-\kappa K}+e^{-\Omega(p)})$; case (β) contributes $E\sum_e\varphi_K^{\beta}(e)\le(C/M+o(1))p$ for every M by Lemma 35(b), i.e. $o(p)$. *Case II:* $T_i^{\mathrm{in}}\ge K$. Clause 1 fails. (II-0) $K\le T_i^{\mathrm{in}}<\varepsilon p$: at most $cpCe^{-\kappa K}$ such edges in expectation (Lemma 32(a) in $G^{\{e\}}$). (II-1) $T_i^{\mathrm{in}}\ge\varepsilon p$, so $i\in\mathcal{T}$. Work on $\Omega'=\Omega\cap\{\max_v \mathrm{indeg}(v)\le C\log p\}$; $P(\Omega'^c)\le Cp^{-2}$, and the edges counted on Ω'^c contribute at most $E[m\mathbf{1}_{\Omega'^c}]\le(Em^2)^{1/2}P(\Omega'^c)^{1/2}=O(1)$. An error here is an edge $e\in E_\uparrow$ with $\Psi_K(e)=0$. Take a witness source node z of (W1), in a source SCC σ. On Ω, $z\in\mathcal{S}$ would give $z\rightsquigarrow i$ (Lemma 33), so $T_z^{\mathrm{out}}<t_0$. (A) $\sigma=\{z\}$ (i.e. z in-isolated) and $T_z^{\mathrm{out}}<K$: clause 2 holds with $s=z$; no error. (B1) z in-isolated and $K\le T_z^{\mathrm{out}}<t_0$. The number of such edges is at most $\sum_z\mathbf{1}[K\le T_z^{\mathrm{out}}<t_0]\sum_{j\in\mathrm{out}(z)}\mathrm{indeg}(j)$. Given $\mathrm{out}(z)=O$ with $|O|=r$ (a completed forward exploration), the pairs with tail outside O and head in O are unrevealed, so the expected number of edges into O is at most $r^2+rpq=r^2+cr$. With $P(T_z^{\mathrm{out}}=r)\le Ce^{-\kappa r}$ (Lemma 32(a)), the expectation is at most $p\sum_{r\ge K}Ce^{-\kappa r}(r^2+cr)\le C'pK^2e^{-\kappa K}$. (B2) $|\sigma|\ge 2$. Then $|\sigma|<t_0$ (as $\sigma\subseteq\mathrm{out}(z)$), every node of σ has out-set $\mathrm{out}(z)$ of size $<t_0$, and the number of edges with head in $\mathrm{out}(z)$ is at most $t_0\cdot C\log p$ on Ω'. Let N_σ be the number of source SCCs with $2\le|\sigma|<t_0$. For a set R of size r, "R strongly connected" and "no edge enters R" depend on disjoint sets of pairs; the first requires every node of R to have a parent in R (probability $\le(rq)^r$, distinct heads), the second has probability $(1-q)^{r(p-r)}$. Hence $EN_\sigma\le\sum_{r=2}^{t_0}\binom{p}{r}(rq)^r(1-q)^{r(p-r)}\le\sum_{r\ge2}\left(ce^{1-c}e^{cr/p}\right)^r=O(1)$, using $\binom{p}{r}\le p^r/r!$, $r^r/r!\le e^r$, $r\le t_0=O(\log p)$ and $ce^{1-c}<1$ for $c\neq1$. So the (B2) errors have expectation $O(1)\cdot t_0C\log p=O(\log^2 p)=o(p)$: an expectation on the event Ω' plus the Cauchy–Schwarz bound for Ω'^c above. Adding all cases gives the claim. □

Theorem 39 (S, $c>1$)**.** *For $c>1$, $|E_\uparrow|/p\to cF_\uparrow$, and therefore $S=\frac12+|E_\uparrow|/(2(p+m))\to\frac12+cF_\uparrow/(2(1+c))$, in L^1 and in probability.*

Proof. $E\big||E_\uparrow|/p-cF_\uparrow\big|\le E\sum_e|\mathbf{1}[e\in E_\uparrow]-\Psi_K(e)|/p+E|\Phi_p^{\Psi_K}-cE_{\mathcal{G}}\Psi_K|+c|E_{\mathcal{G}}\Psi_K-F_\uparrow|$; the terms are bounded by Prop. 38, Lemma 35(a) and Prop. 37; let $p\to\infty$, then $K\to\infty$. $m/p\to c$ in L^2 ($m\sim\mathrm{Bin}(p(p-1),q)$), and S and L are bounded continuous functions of $(|E_\uparrow|/p,m/p)$ and $(N/p,m/p)$ on $[0,\infty)^2$. □

Proposition 40 ($c\le1$)**.** *(a) For $c<1$, $EN\le c^2/(1-c)$. (b) For $c\le1$, $N/p\to0$ in L^1, so $L\to\frac12+\frac{c}{2(1+c)}$ ($L(1)=3/4$). (c) For $c\le1$, $|E_\uparrow|/p\to c(1-e^{-c})$ in L^1, so $S\to\frac12+\frac{c(1-e^{-c})}{2(1+c)}$ ($S(1)=0.65803$); this is $F_\uparrow$ at $x=0$.*

Proof. (a) A non-loop edge inside an SCC lies on a simple directed cycle, and the expected number of pairs (simple cycle of length k, edge on it) is $k\cdot\frac{(p)_k}{k}q^k\le c^k$; sum over $k\ge2$. (b) $c<1$ from (a). $c=1$: $D(p,1/p)$ is a subgraph of $D(p,c'/p)$ for every $c'>1$ under the monotone coupling, and an edge inside an SCC of the subgraph is inside an SCC of the larger graph, so $\limsup_p EN/p\le c'x(c')^2\to0$ as $c'\downarrow1$ (Prop. 36). (c) Let $\Psi_K^1=\mathrm{cl}_1$. Case I ($T_i^{\mathrm{in}}<K$): (I-a) as before, no error; (I-b) with the same union over other parents k and forced $F=\{(i,j),(k,j)\}$, (W3) forces $\mathrm{in}(k)\cap\mathrm{in}(i)\neq\emptyset$, so either $T_k^{\mathrm{in}}\ge K$ or the event of φ_K^{β} holds. The expected number of triples with $T_k^{\mathrm{in}}\ge K$ is $E\sum_e\#\{k\notin\{i,j\}:k\to j,\ T_k^{\mathrm{in}}\ge K\}$, a local functional dominated by $\mathrm{indeg}(j)$, so by Lemma 35(b) it is at most $(c\cdot c\theta_K+C/M+o(1))p$; the φ_K^{β} part is $o(p)$ as before. Case II ($T_i^{\mathrm{in}}\ge K$): at most $E\sum_e\mathbf{1}[T_i^{\mathrm{in}}\ge K]=(c\theta_K+o(1))p$ (Lemma 35(a)). Hence $\limsup_p E\sum_e|\mathbf{1}[e\in E_\uparrow]-\Psi_K^1(e)|/p\le c^2\theta_K+c\theta_K+C/M$, which tends to 0 as $K,M\to\infty$ because $\theta_K\to x(c)=0$ for $c\le1$ (F3). Finally $E_{\mathcal{G}}\Psi_K^1=(1-\theta_K)(1-e^{-c})\to1-e^{-c}$. □

What is not proved. Rates of convergence; the critical window $c=1+\lambda p^{-1/3}$ (Łuczak and Seierstad, recorded in Dovgal et al., 2020; not used); almost sure convergence; the regime of fixed T with $p\to\infty$. The proof does not use the structure theorems of Karp and of Łuczak (not opened by us); it re-proves the part it needs.

Numerical checks of the intermediate claims (fixed seeds, exact computation; supplement `x1proof/verify`). (W4) $E_\uparrow\subseteq$ edges into confluences: 0 violations in 48 graphs. Dichotomy (Lemma 32): at $c=2$, $p=10^5$, the mass of $K\le T<0.02p$ is 0.0097, 0.0008, 0 for $K=5,10,20$; at $c=1$ it is 0.37, 0.25, 0.17 (critical, no dichotomy, as expected). Structure (Corollary 34): for $c\ge1.5$ the second-largest SCC never had more than 2 nodes; giant fraction 0.63439 ($p=10^6$, $c=2$) against $x^2=0.63491$. Prop. 38 edge by edge: at $c=2$, $p=2\cdot10^5$, $|E_\uparrow|=76{,}220$ and $\sum_e\Psi_{40}(e)=76{,}220$ with no edge in either difference; clause 2 never erred in any run. Concentration: the standard deviation of N/p times $\sqrt{p}$ is 2.88–3.07 for $p=5\cdot10^3$ to $3.2\cdot10^5$ ($c=2$), as Lemma 35 requires.

O. Accuracy Interpretation of s_ρ

An edge $\{x, y\}$ of a DAG is shown in a uniformly random order together with $(\rho_{\text{first}}, \rho_{\text{second}})$. A *pairwise count rule* is $f(a, b) \in [0, 1]$, the probability of declaring "first is the tail"; the *sort rule* is $f_{\text{sort}} = 1, \frac{1}{2}, 0$ for $a <, =, > b$. For the true edge $x \to y$ the accuracy is $\frac{1}{2} f(\rho_x, \rho_y) + \frac{1}{2}(1 - f(\rho_y, \rho_x))$.

Proposition 41 (elementary)**.** *On every DAG, (i) the accuracy of f_{sort} is exactly s_ρ; (ii) every pairwise count rule has accuracy at most s_ρ; (iii) f_{sort} never errs on an edge with $\rho_x \neq \rho_y$.*

Lean: `accRule_sortRule`, `accRule_sortRule_rho`, `accRule_le_sOf`, `accRule_le_sRho`, `edgeAcc_sortRule_strict`, `rho_not_gt_of_edge`.

Proof. By Reisach et al.'s Theorem 1, $\rho_x \leq \rho_y$ on an edge $x \to y$. If $\rho_x < \rho_y$ the accuracy is at most 1, attained by the sort rule; if $\rho_x = \rho_y = a$ it is $\frac{1}{2} f(a, a) + \frac{1}{2}(1 - f(a, a)) = \frac{1}{2}$ for every f, because the random order forces the tie decision to $1/2$. These are the edge scores. (i) restates the definition of s_ρ; (ii) is a three-line consequence of their Theorem 1. □

Combined with Theorem 5, the best accuracy of any pairwise count rule on the stationary unrolled DAG is $\frac{1}{2} + |E_\uparrow|/(2|E_S|)$, at most 0.665 for ER summary graphs and 0.75 for river-like in-forests ($\mu \leq 1$). The restriction to counts is necessary (exhaustive enumeration over labelled DAGs, not proved):

nodes p	2	3	4	5
Bayes rule on the two counts (= edge-pooled s_ρ)	0.5	0.5625	0.592262	0.605547
Bayes rule on counts and the two node degrees	0.5	0.625	0.660714	0.669238

For the skeleton $a - c - b$ the collider has counts $(2, 2, 3)$ (all edges strict) while the chain and the fork have all counts equal to 3; among the tied graphs the centre c is the tail with probability $2/3$ under a uniform prior, so a degree-aware rule beats the coin. So s_ρ is the Bayes accuracy of pairwise count rules only. This section is interpretive, not a new theorem.

P. Benchmark Experiment B1: Design, Results and Deviations

Process. The analysis plan (`b1/PLAN.md`) was committed before any simulation that produced an algorithm score; only graph-only exact values were computed before it. Two deviations were recorded in the plan after the validation run (seeds 9000–9004, used only for debugging and timing) and before the test run: D1, the burn-in was raised from 50 to 100 steps because the validation mixing check failed (error $2.8 \cdot 10^{-3}$, threshold 10^{-3}; $1.4 \cdot 10^{-10}$ after the change); D2, SID is undefined when PC returns a cyclic directed part, so PC is dropped from the SID comparison in those cells (930 instead of 1,190 tests). D3: a container restart killed the first test run after about half of one shard; no results file had been written and no partial score was inspected. The test run was redone from scratch with the same deterministic seeds. No threshold or hypothesis was changed. As with the other studies, the plan is internally time-stamped and agent-executed (Appendix I).

Design. Summary graphs with a self-loop at every node, $p \in \{5, 10\}$, 30 test graphs (seeds 1000–1029) per family: symmetrised ER (expected degree 2) and the directed ring (strongly connected), the chain, random recursive out- and in-trees, and directed ER $G(p, c/p)$ with $c \in \{1, 2\}$. Linear Gaussian VAR(1) with every weight $\sim \text{Unif}(0.5, 1)$, rescaled to spectral radius 0.8 if larger, so relatives in the population equal graph relatives; noise s.d. $\sim \text{Unif}(0.5, 1)$ per node (R-het, primary), equal noise (R-eq), and R-het standardised (R-std). G1: x_1 is noise, then $T - 1$ steps, truth G_T. G2: 100 latent burn-in steps, then the window of T slices, truth the window copy of G_T; the burn-in makes slice-B nodes latent common parents of the first window slice, so we also report SHD excluding pairs inside the first slice. G2s: exact stationary start via the discrete Lyapunov equation (replication of G2). Methods with columns shuffled: rel-SortnRegress (Reisach et al. Algorithms 1–2, $\alpha = 0.05$), R^2-SortnRegress, Var-SortnRegress, a random order followed by the same regression step, FLOP (`flopsearch` 0.3.0, BIC penalty 2, 10 restarts), PC (causal-learn, Fisher z, $\alpha = 0.01$; not run at $p = 10, T = 10$ for time), and a labels-visible reference that sorts by time. DAGMA/NOTEARS were not run. SID by `gadjid`.

Pre-specified hypotheses and verdicts. H1 (start helps rel-SortnRegress on non-strongly-connected families; at least 4/5 Holm-significant and pooled mean ≥ 0.02): supported, 5/5, pooled $+0.125$ $[0.116, 0.134]$; tracking (Spearman of cell means against the exact gap ≥ 0.5): 0.80, $p = 1.2 \cdot 10^{-10}$; OLS $d = 0.53\,\text{gap} - 0.027$. H2a (gain below 0.03 in every strongly connected cell with $T \geq p$, the review's expectation): fails, as the exact gaps predicted before the run (e.g. ring $p = 5, T = 5$: $+0.217$; only symER $p = 5, T = 10$ is small, $+0.019$). H2b (non-negligible gain at $T < p$): holds, $+0.13$

to $+0.31$ in all six cells. H2c (symER gain decreases with T): holds, $+0.195$, $p = 1.2 \cdot 10^{-11}$. H3 (some Holm-significant SHD rank swap): supported, 11 swaps: rel- vs R^2-SortnRegress on chains ($p = 10$, $T = 3, 5$), Var-SortnRegress vs FLOP (3 cells) and vs PC (5 cells) on in-trees, and R^2-SortnRegress vs the random order on one ring cell. SID: 9 swaps; orientation score: 9; SHD without first-slice pairs: 9. Median Kendall τ between the G1 and G2 rankings: 0.73. FLOP is best on average under both generators (normalised SHD 0.27 vs 0.46).

Two artefacts. Pooled G1$-$G2 order accuracy: rel-SortnRegress $+0.133$ (unchanged by standardisation, since it is scale invariant); Var-SortnRegress $+0.202$ raw, $+0.244$ with equal noise, $+0.002$ standardised. Variance accumulates from the noise-only first slice, so the fixed start is var-sortable; standardising removes this but not the relatives artefact. Only the rel- vs R^2-SortnRegress swaps follow the mechanism of Theorems 2 and 5.

Cost of the burn-in. The labels-visible reference has normalised SHD 0.044 (G1) and 0.273 (G2), but 0.041 vs 0.048 without first-slice pairs (G2s identical): stationary-start benchmarks should score the window without first-slice pairs or as an ADMG.

Exact curves and the Δ bound. Figure 4 shows exact $s_\rho(G_T)$ for $T = 2, \ldots, 20$. At $T = 3$ it lies between 0.81 and 1.00; at $T = 20$ symER is still at 0.55–0.57 and the ring with $p = 10$ at 0.74. The Δ form of Theorem 2 was checked against brute-force relatives enumeration on 3,000 random summary graphs $\times$ 7 horizons: no violation of either bound, equality in 3,164 cases, strictly sharper than the $p - 1$ bound in 10,186. For Theorem 5, $B = \Delta$ gave the stationary value in 726 of 726 graphs and $B = \Delta - 1$ did not in 445, so Δ cannot be lowered in general.

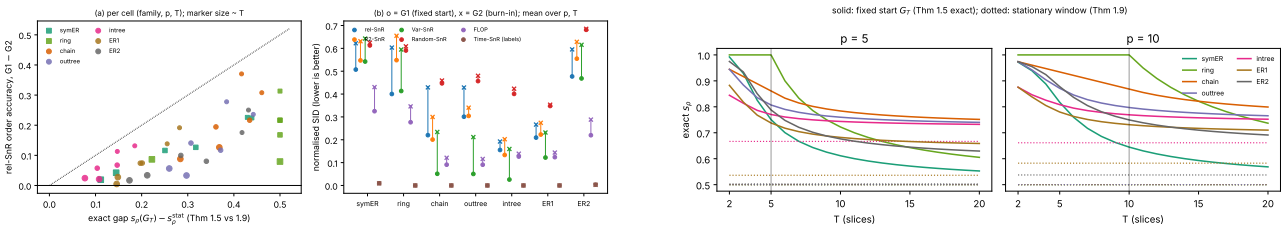


Figure 4. Left: per-cell G1$-$G2 gain in rel-SortnRegress order accuracy against the exact population gap (all 42 cells below the diagonal and above 0), and normalised SID per method under G1 and G2. Right: exact $s_\rho(G_T)$ against T, mean over the 30 test graphs per family and p.